MILITARY SPECIFICATION

TEST SET, IFF INTERROGATOR AN/TPM-25A

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers one type test set which is used in conjuction with the AN/USM-281 Oscilloscope, or equivalent, and the AN/TPM-24(V) Test Set, Test Facilities Set to check, maintain, and calibrate IFF (Identification-Friend-or-Foe) Interrogators and selected IFF Test Sets. The test set covered by this specification is designated Test Set, Radar AN/TPM-25A.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)
PPP-P-291	Paperboard, wrapping, Cushioning
PPP-F-320	Fiberboard, Corrugated and solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
QQ-S-781	Strapping, Steel, Flat and seals

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MIL-P-116	Preservation, Packaging Methods of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: DRSEL-ED-TO, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-B-117	Bag, Sleeve and Tubing-Interior Packaging
MIL-P-11268	Parts, Materials, and Processes used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam Expanded Bead)
MIL-T-21200	Test Equipment for use with Electronic and Electrical Equipment, General Specification for
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-P-55110	Printed Wiring Boards
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging of

DOD

DOD-AIMS 65-1000 (Classified-CONFIDENTIAL)	Technical Standard for the ATCRBS/IFF/Mark XII System

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables of Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electrical Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods

US NATIONAL AVIATION STANDARD

1010.51A	IFF Mark X(SIF) Air Traffic Control Radar Beacon System (ATCRBS) Characteristics

INSTRUCTIONS

SPI 1G00118	Special Packaging Instructions

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procurement activity or as directed by the Contracting Officer).

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3. REQUIREMENTS.

3.1 DESCRIPTION. The Test Set, Radar AN/TPM-25A consists of a Test Set, TS-3394/TPM-25A with a removable cover and associated cable assemblies, accessory attenuators, connector adaptors, terminations, Video Probe, and Conversion Chart. The TS-3394/TPM-25A provides stimuli and measurement capabilities required for shop maintenance of IFF interrogators. The test set checks operational characteristics, such as receiver bandpass characteristics and sensitivity, decoding and coding of all modes, Transmitter output power and frequency, and receiver sidelobe suppression. The TS-3394/TPM-25A operates from an external 115 VAC power source.

3.2 FIRST ARTICLE. Specified in the contract, the contractor shall furnish first article units (See 4.3).

3.3 GENERAL REQUIREMENTS. The AN/TPM-25A shall be in accordance with MIL-T-21200, Class 3, unless otherwise specified herein (See 4.4).

3.4 PARTS, MATERIALS AND PROCESSES. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply (See 4.4).

3.4.1 GENERAL.

a. When non-pressurized castings are used, the requirement for impregnation as stated in MIL-STD-276 shall not apply.

b. When a soft pliable coating (polyurethane) per MIL-I-46058 is used for conformal coating, the third and fourth sentences of paragraph 3.10.31.1.1 shall not apply.

c. When anti-sieze compounds may affect RF grounding, the requirement for their use shall not apply.

3.5 FINISH, PROTECTIVE. The AN/TPM-25A shall be given protective finish in accordance with Specification MIL-F-14072. This includes primer and finish of hardware, such as handles, hinges, screws, etc., and necessary touch up after mounting. The final paint film on Type I surfaces shall be final Film E, semigloss, light-gray enamel, conforming to Specification MIL-F-14072 (See 4.4)

3.6 MARKING.

3.6.1 GENERAL. Marking shall conform to Specification MIL-M-13231. Front panel marking shall be Group I, as described in that specification (See 4.4).

3.6.2. VISIBILITY. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment (See 4.4).

3.6.3 SERIAL NUMBERS. The following items shall have serial numbers:

a. Test Set, AN/TPM-25A.

b. Test Set, TS-3394/TPM-25A (See 4.4).

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3.7 FRONT PANEL PROTECTION. Handles shall be provided on the front of the TS-3394/TPM-25A hereafter known as the equipment. The outer edge of the handles shall extend at least the distance of the controls, indicator or electrical connector indicator or electrical connector that has the greatest protrusion from the front panel (See 4.6).

3.8 INPUT POWER SOURCE. The equipment shall be capable of operating from a 115 VAC, ±10%, 60/400 Hz power source in accordance with MIL-T-21200, except the low frequency limit for the power source shall be 45 Hz (See 4.7).

3.8.1 POWER SWITCH. A two position toggle switch, identified as "Power, ON-OFF", shall permit the equipment to be energized by the 115 VAC power source (See 4.7.1).

3.8.2 PROTECTION, ELECTRICAL. Electrical overload protection shall be in accordance with Requirment 8 of MIL-STD-454. Fuses shall be of the indicating type and shall be in accordance with Requirement 39 of MIL-STD-454.

3.8.3 POWER INDICATOR. AN AC power indicator lamp and lamp holder with lens, shall be provided on the front panel of the equipment, to indicate the application of the 115 VAC power source. The indication shall be green in color (See 4.7.2).

3.9 EXTENDER CARD. An extender card shall be provided for the equipment printed circuit cards and shall be keyed to prevent improper insertion of itself or the circuit cards. The extender card shall be stored within the AN/TPM-25A in a location satisfactory to the Government (See 4.8).

3.10 INSTRUMENT CASE. An instrument case shall be provided to house the equipment chassis. A Cover, Test Set MX-9488/TPM-25A shall be provided with the instrument case for transportability (See 4.20). Any requirement for projecting feet for stacking purposes shall not apply.

3.10.1 ORIENTATION. The instrument case shall permit installation of the equipment chassis within the case in only one position (See 4.20).

3.10.2 CHASSIS FASTENERS. The equipment shall be contained within the instrument case by means of captive screws that permit secure fastening and ease in removal (See 4.20). Screws which are used to hold internal parts (switches, knobs, etc) need not be a similar color as the chassis panel.

3.10.3 HANDLES. The instrument case shall have two handles, one at each end, optimally located to facilitate portability of the equipment (See 4.20) and in accordance with MIL-STD-1472B.

3.10.4 COVER. The instrument case cover, Cover, Test Set MX-9488/TPM-25A shall protect the front panel controls and indicators and provide storage of the equipment accessories (See 4.20). A schematic diagram and brief operating instructions are not required to be affixed to the inside of the cover.

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3.11 WEIGHT The weight of the equipment, including the instrument case cover and accessories shall not exceed 70 pounds (See 4.10).

3.12 VOLUME. The volume of the equipment, including the instrument case with cover, shall not exceed 2.7 cubic feet. The form factor shall be such that the front panel dimensions shall not exceed 15 inches by 19 inches (See 4.11).

3.13 GOVERNMENT FURNISHED PROPERTY. Government furnished property will be furnished to the contractor as specified in the contract.

3.14 COMPATIBILITY. (See 4.12).

3.14.1 IFF SYSTEM COMPATIBILITY. The equipment shall be compatible with the IFF Mark X(SIF)/Air Traffic Control Radar Beacon System SIF/ATCRBS, in accordance with U.S. National Aviation Standard 1010.51A and shall be compatible with the characteristics of Specification DOD-AIMS 65-1000 (See 4.12).

3.14.2 EQUIPMENT COMPATIBILITY. The equipment, when used in conjunction with the Test Set, Test Facilities Set AN/TPM-24 and the Oscilloscope AN/USM-281, or equivalent, shall be operationally compatible with the Interrogator Sets AN/TPX-46 and AN/TPX-50 and Test Set, Transponder AN/APM-123 (See 4.12).

3.14.3 CHASSIS GROUND. A separate front panel terminal, shall be provided to permit grounding the equipment chassis to the chassis of the associated equipment (4.12).

3.15 RELIABILITY. The equipment shall have a mean-time-between-failure (MTBF) of 900 hours when subjected to the environmental conditions of sec. 4.5.5.

3.16 ELECTROMAGNETIC INTERFERENCE EMISSION AND SUSCEPTIBILITY. The equipment shall meet the following requirements of MIL-STD-461A, Notice 4 (See 4.13):

CE02	CS02	RE02	RS02
CE04	CS06	RE02.1	RS03*
			RS03.1*

*The limits for nonsheltered equipment shall apply.

3.17 WARM-UP TIME. The equipment warm-up time shall not exceed ten minutes when operating at room ambient and at extreme operating temperature conditions (See 4.14).

3.18 SERVICE CONDITIONS. The equipment shall conform to the requirements of this specification for the following service conditions (See 4.15).

3.18.1 TEMPERATURE.

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3.18.1.1 HIGH TEMPERATURE. The equipment shall meet the specified performance requirements of Table IV after storage at 85°C and when operated at 55°C. The time duration of high temperature storage at 85°C shall be 48 hours (See 4.15.1.1).

3.18.1.2 LOW TEMPERATURE. The equipment shall meet the specified performance requirements of Table IV after storage at -62°C and when operated at 0 C. The time duration of low temperature storage at -62° C shall be 4 hours (See 4.15.1.2).

3.18.2 ALTITUDE. The equipment shall meet the specified performance requirements of Table IV when operating up to 10,000 feet above sea level and withstand, in an non-operation condition, exposure up to 50,000 feet above sea level (See 4.15.2).

3.18.3 HUMIDITY. The equipment shall withstand the effect of humidities up to 100 percent, including conditions wherein condensation takes place in and on the equipment. The equipment shall withstand the above conditions during operating and non-operating conditions (See 4.15.3).

3.18.4 VIBRATION. The equipment in a non-operating condition shall be capable of withstanding vibration encountered during service conditions (See 4.15.4).

3.18.5 SHOCK.

3.18.5.1 BENCH HANDLING. The equipment shall be capable of withstanding shock encountered in servicing (See 4.15.5.1).

3.18.5.2 BASIC DESIGN. The equipment shall be capable of withstanding shock induced by routine handling (See 4.15.5.2).

3.18.6 FUNGUS. The equipment shall be resistant to fungus growth (See 4.15.6).

3.18.7 SALT FOG. The equipment shall be capable of withstanding exposure in normal operating configuration, to salt fog and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish, or clogging or binding of moving parts (See 4.15.7).

3.19 ELECTRICAL REQUIREMENTS, TEST SET, RADAR TS-3394/TPM-25A. Unless otherwise specified, the equipment shall be capable of meeting all the following requirements (See 4.16):

3.19.1 PULSE REPETITION FREQUENCY(PRF). A three-position front panel switch shall be provided to select an internal trigger, external trigger or decode trigger to control the equipment PRF (See 4.20).

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3.19.1.1 INTERNAL TRIGGERED PRF. Four front panel switches shall select the period of the equipment's PRF for internal trigger operation from 500 USEC to 9,999 USEC in one microsecond steps (See 4.16.1)

3.19.1.1.1 ACCURACY. The PRF period shall be within ±0.2 USEC or ±0.01% of the indicated setting, whichever tolerance is greater (See 4.16.1.1).

3.19.1.1.2 JITTER. The period-to-period jitter shall not exceed ±0.06 USEC (See 4.16.1.2).

3.19.1.2 EXTERNAL TRIGGERED PRF. The equipment's PRF for external trigger operation shall be controlled by an external trigger having the following characteristics when terminated in 75 ohms ±10% (See 4.16.2):

a.	Polarity	Positive or negative
b.	Amplitude	5 volts to 25 volts
c.	Width	0.25 USEC to 5.0 USEC
d.	Rise Time	0.1 USEC or less
e.	Decay Time	1.0 USEC or less
f.	Repetition Rate	101 Hz to 2000 Hz

3.19.1.2.1 PRF GENERATION. One PRF cycle shall be generated for each external trigger (See 4.16.2.1).

3.19.1.3 DECODE TRIGGERED PRF. For fixed frequency operation only (3.19.14.4.1), the equipment shall decode Modes 1, 2, 3/A, 4 and C RF challenges having repetition rates of between 101 Hz and 2000 Hz and provide an internal decode trigger to control the equipment's operating PRF (See 4.16.3).

3.19.1.3.1 PRF GENERATION. One equipment PRF cycle shall be generated for each challenge decode (See 4.16.3.1).

3.19.1.3.2 MODE SELECTION. A panel switch shall be provided to select the challenge mode to be decoded (See 4.16.3.2). This switch shall enable the equipment's decode capability (See 4.16.3.3) when any challenge mode is selected.

3.19.1.4 SWEEP FREQUENCY MODE PRF. For the sweep modes of operation, the equipment shall be capable of operating at the following PRF's (See 4.16.4):

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SWEEP BANDWIDTH	PRF RANGE
Narrowband (±5 MHz)	101 Hz to 800 Hz ±20%
Wideband (±15 MHz)	101 Hz to 450 Hz ±20%

3.19.1.4.1 PRF COUNTDOWN. For selected PRF's exceeding the maximum limits (3.19.1.4), the equipment shall automatically countdown and select the largest sub-multiple repetition rate which is below 800 Hz ±20% for the ±5 MHz bandwidth and below 450 ±20% for the ±15 MHz bandwidth (see 4.16.4.1).

3.19.2 RADAR TRIGGER. The equipment shall generate, except for decode-triggered PRF operation (3.19.1.3), a radar trigger with the leading edge used as a zero time reference (See 4.16.5).

3.19.3 RADAR TRIGGER CHARACTERISTICS. The radar trigger output shall be provided at the front panel for all modes of operation, with the following characteristics when terminated in 75 ohms ±10% (See 4.16.6):

a.	Amplitude	10.0 ±1.0 volts positive
b.	Width	3.0 ±1.0 USEC
c.	Rise Time	0.1 USEC or less
d.	Decay Time	0.2 USEC or less
e.	Timing	Within 0.3 USEC of the external Trigger for external triggered PRF's.

3.19.4 SELECTIVE IDENTIFICATION FEATURE (SIF) PRETRIGGER. An SIF pretrigger output shall be provided at the front panel for all modes of operation, except decode triggered PRF (3.19.1.3), with the following characteristics when terminated in 75 ohms ±10% (See 4.16.7).

a.	Amplitude	20.0 ±4.0 volts positive
b.	Width	0.8 USEC ±0.1 USEC
c.	Rise Time	0.1 USEC or less
d.	Decay Time	0.2 USEC or less
e.	Timing	386.0 ±.5 USEC after the Radar trigger

3.19.4.1 ADJUSTMENT. Internal controls shall be provided to adjust the pretrigger timing in 0.2 USEC steps from 301.0 to 398.0 USEC after the radar trigger (See 4.16.7.1).

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3.19.5 MODE 4 GAIN TIME CONTROL(GTC) TRIGGER. A Mode 4 GTC trigger output shall be provided at the front panel, with the following characteristics when terminated in 90 ohms ±10% (See 4.16.8).

a.	Amplitude	5.0 volts ±0.5 volt positive
b.	Width	0.80 ±0.2 USEC
c.	Rise Time	0.10 USEC or less
d.	Decay Time	0.2 USEC or less
e.	Jitter	Less than +0.5 USEC with respect to the radar trigger
f.	Timing	372.0 ±0.4 USEC after the Mode 4 internal reference trigger (3.19.8.3).

3.19.6 SCOPE TRIGGERS. For all modes of operation, an output trigger shall be provided at the front panel to trigger the Oscilloscope AN/USM-281 or equivalent. The output trigger pulse shall have the following characteristics when terminated in 90 ohms ±10% (See 4.16.9):

a.	Amplitude	5.0 ±1.0 volts
b.	Rise Time	0.1 USEC or less
c.	Decay Time	0.5 USEC or less
d.	Width	1.0 USEC to 5.0 USEC

3.19.6.1 FIXED FREQUENCY SCOPE TRIGGER. The scope trigger shall occur within 0.50 USEC after the radar trigger (3.19.3) for fixed frequency operation (See 4.16.9.1).

3.19.6.1.1 JITTER. Jitter shall not exceed ±0.05 USEC with respect to the radar trigger (See 4.16.9.1.1).

3.19.6.2 SWEEP SYNC SCOPE TRIGGER. For sweep frequency operation, a sweep sync output trigger shall be provided within ±1 USEC of the start of each sweep (See 4.16.9.2).

3.19.6.3 DECODE SCOPE TRIGGER. The scope trigger shall occur within ±1.0 USEC of the last challenge pulse for SIF operation and 193 ±3 USEC after the fourth sync pulse of a Mode 4 challenge (See 4.16.9.3).

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3.19.7 DELAYED RF SWEEP TRIGGER. For sweep frequency operation the start of the sweep cycle, with respect to the radar trigger, shall be continuously variable, by means of a front panel control, from less than 10 USEC to greater than 10,000 USEC (See 4.16.9.4).

3.19.8 SYSTEM TIMING.

3.19.8.1 REPLY TIMING DELAY RANGE. For internally, externally or decode-generated PRF operation, the equipment's coded video replies shall be capable of being delayed over a range of 1 USEC to 2998 USEC (See 4.16.10.1).

3.19.8.1.1 DELAY SELECTION. The amount of delay shall be selectable in 1 USEC increments using four front panel switches with an accuracy of ±0.2 USEC for delays between 1 USEC to 999 USEC and ±0.6 USEC for delays between 1000 USEC to 2998 USEC (See 4.16.10.1.1).

3.19.8.2 SIF SYSTEM TIMING.

3.19.8.2.1 TRIGGERED OPERATION. For SIF internal or external trigger operation, the minimum setting (1.0 USEC) of the reply delay switches (3.19.8.1.1) shall cause the first bracket pulse of the SIF reply (3.19.12.1) to occur at a nominal position of 434.0 ±0.5 USEC after the radar trigger (See 4.16.10.2.1).

3.19.8.2.1.1 REPLY TIMING ADJUSTMENT. Internal controls shall be provided to position the occurrence of the first bracket pulse, of the earliest SIF reply, in 0.1 USEC steps, from 360.0 USEC to 438 USEC after the radar trigger with the range delay switches set for minimum setting (See 4.16.10.2.1.1).

3.19.8.2.2 DECODE OPERATION. For decode trigger operation, the earliest occurrence of an SIF reply shall be 4.0 ±1.0 USEC after the last SIF challenge pulse (3.19.11) (See 4.16.10.2.2).

3.19.8.3 MODE 4 SYSTEM TIMING.

3.19.8.3.1 TRIGGERED OPERATION. For Mode 4 internal or external trigger operation, an internal reference trigger shall be generated at a nominal position of 14.5 ±0.2 USEC after the radar trigger to control the Mode 4 system timing (See 4.16.10.3.1).

3.19.18.3.1.1 REFERENCE TRIGGER TIMING ADJUSTMENT. Internal controls shall be provided to position the occurrence of the reference trigger, in 0.1 USEC steps, from 4.0 USEC to 34 USEC after the radar trigger (See 4.16.10.3.1.1).

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3.19.8.3.1.2 TIMING ADJUSTMENT SYNCHRONIZATION. Variations in the positioning of the reference trigger shall automatically cause corresponding variations in the positioning of the equipment's Mode 4 replies (3.19.12.2) GTC trigger (3.19.5) and challenge video (3.19.10) (See 4.16.10.3.1.2).

3.19.8.3.1.3 REPLY POSITION. The first pulse of a three-pulse Mode 4 reply (3.19.12.2.1) shall occur 434.5 ±0.4 uSEC after the internal reference trigger with the reference trigger set to its nominal position and the reply delay switches set for minimum delay (See 4.16.10.3.1.3).

3.19.8.3.2 DECODE OPERATION. For decode trigger operation, the earliest occurrence of the first pulse of a three-pulse Mode 4 reply shall be 260.5 ±3.0 USEC after the fourth challenge sync pulse (3.19.10) (See 4.16.10.3.2).

3.19.9 TIMING INTERLOCK. The equipment shall contain a timing interlock capability to inhibit reply outputs when the equipment pulse repetition time (PRT) period is less than the time from the radar trigger to the first F2 pulse (3.19.12.1.1) of an SIF reply or the last pulse of a Mode 4 reply (See 4.16.11).

3.19.9.1 INDICATOR. A front panel indicator shall illuminate when the equipment reply output is inhibited (See 4.16.11.1).

3.19.10 MODE 4 CHALLENGE VIDEO OUTPUT. An output pulse train shall be generated consisting of 37 pulses spaced at multiples of 2.0 USEC from the first pulse with the exception that the fifth pulse position (8.0 USEC) shall be vacant (See 4.16.12).

3.19.10.1 PULSE TRAIN TIMING. The first challenge video pulse shall occur 168.0 ±0.4 USEC after the Mode 4 internal reference trigger (3.19.8.3)(See 4.16.12.1).

3.19.10.2 JITTER. The jitter between the first challenge video pulse and the radar trigger shall be less than ±0.05 USEC (See 4.16.12.2).

3.19.10.3 PULSE CHARACTERISTICS. Each challenge video output pulse shall have the following characteristics when terminated in 90 ohms ±10% (See 4.16.12.3):

a.	Amplitude	5.0 ±0.5 volts positive
b.	Width	0.50 ±0.05 USEC

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c.	Rise Time	0.10 USEC or less
d.	Decay Time	0.20 USEC or less
e.	Spacing Accuracy	±0.07 USEC from nominal

3.19.11 SIF CHALLENGE VIDEO OUTPUT. The equipment shall provide an output at the front panel for SIF challenge video pulse pairs with the following characteristics when terminated in 75 ohms ±10% (See 4.16.13):

a.	Amplitude	Selectable as either 5 ±0.5 volts positive or 20.0 ±4.0 volts positive
b.	Width	0.80 ±0.05 USEC
c.	Rise Time	0.10 USEC or less
d.	Decay Time	0.20 USEC or less

3.19.11.1 AMPLITUDE SELECT SWITCH. A two position, front panel toggle switch with an associated switch guard shall be provided to Select the amplitude of the SIF challenge video output. The switch guard shall allow the switch to be set in either of its positions but shall require removal from the front panel to permit the movement of the switch toggle from one position to the other (See 4.20).

3.19.11.2 CHALLENGE VIDEO PULSE SPACING. Front panel controls shall permit the selection of the following puse pair spacings (See 4.16.13.1):

MODE	SPACING
1	3.0 ± 0.1 USEC
2	5.0 ± 0.1 USEC
3/A	8.0 ± 0.1 USEC
C	21.0 ± 0.1 USEC

3.19.11.3 CHALLENGE VIDEO TIMING. The second pulse of each pulse pair shall occur 4.0 ±0.2 USEC before the first bracket pulse of an SIF reply (3.19.12.1) with the range delay switches set for minimum setting (See 4.16.13.2).

3.19.12 VIDEO CHARACTERISTICS. The equipment shall generate encoded IFF video signals as specified herein. Reply selection shall be accomplished using the modulation select switch (3.19.14.9.1) (See 4.20).

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3.19.12.1 SIF REPLIES. The equipment shall provide the replies specified in 3.19.12.1.1 through 3.19.12.1.5.1 at a front panel connector with the following individual pulse characteristics when terminated in 75 ohms ±10% (See 4.16.14.1).

a.	Amplitude	5.0 ±0.5 volts positive
b.	Width (Nominal)	0.45 ±0.05 USEC
c.	Rise Time	0.1 USEC or less
d.	Decay Time	0.2 USEC or less

3.19.12.1.1 SIF NORMAL REPLIES. The normal SIF replies shall consist of a pair of bracket pulses, F1 and F2, plus a maximum of twelve information pulses having the following spacings (See 4.16.14.1.1):

PULSE	SPACING
F1 (Bracket 1)	0 USEC (Reference)
C1	1.45 ±0.02 USEC
A1	2.90 ±0.02 USEC
C2	4.35 ±0.02 USEC
A2	5.80 ±0.02 USEC
C4	7.25 ±0.02 USEC
A4	8.70 ±0.02 USEC
B1	11.60 ±0.02 USEC
D1	13.05 ±0.02 USEC
B2	14.50 ±0.02 USEC
D2	15.95 ±0.02 USEC
B4	17.40 ±0.02 USEC
D4	18.85 ±0.02 USEC
F2 (Bracket 2)	20.30 ±0.02 USEC

3.19.12.1.1.1 CODE SELECT. The availability of the information pulses shall be controlled by four front panel code select switches for SIF code selection of 0000 through 7777 (See 4.16.14.1.2).

3.19.12.1.2 SIF IDENTIFICATION/POSITION(I/P) MODE 1 REPLY. The reply shall consist of two SIF pulse trains as specified in 3.19.12.1.1 with a spacing of 24.65 ±0.05 USEC between the Bracket 1 pulses of each train (See 4.16.14.1.3).

3.19.12.1.3 SIF I/P MODE 2/3 REPLY. The reply shall consist of a single SIF pulse train (3.19.12.1.1) followed by a pulse spaced 24.65 ±0.05 USEC after the Bracket 1 pulse (See 4.16.14.1.4).

3.19.12.1.4 SIF EMERGENCY REPLY. The reply shall consist of a single SIF pulse train (3.19.12.1.1) followed by three sets of Bracket pulses spaced as follows with respect to the first Bracket pulse of the initial pulse train (See 4.16.14.1.5):

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Bracket 1, Set 1	24.65 ±0.05 USEC
Bracket 2, Set 1	44.95 ±0.1 USEC
Bracket 1, Set 2	49.30 ±0.1 USEC
Bracket 2, Set 2	69.60 ±0.1 USEC
Bracket 1, Set 3	73.95 ±0.1 USEC
Bracket 2, Set 3	94.25 ±0.1 USEC

3.19.12.1.5 SIF GARBLE REPLY. The reply shall consist of an SIF pulse train (3.19.12.1.1) and a single garble pulse spaced 20.3 ±0.05 USEC after the information pulse selected as the substitute pulse (3.19.12.4) (See 4.16.14.1.6).

3.19.12.1.5.1 GARBLE VARIABLE PULSE SPACING. The substitute pulse spacing control of 3.19.12.4.1 shall control the position of the substitute pulse and the delayed garble pulse to maintain a spacing of 20.3 ±0.05 USEC (See 4.16.14.1.6.1).

3.19.12.2 MODE 4 REPLIES. The equipment shall provide either single pulse or three pulse replies at a front panel connector with the following individual pulse characteristics when terminated in 75 ohms ±10% (See 4.16.14.2):

a. Amplitude	5.0 ±.5 volts positive
b. Width	0.5 ±0.1 USEC
c. Rise Time	0.1 USEC or less
d. Decay Time	0.2 USEC or less

3.19.12.2.1 THREE-PULSE REPLY SPACING. The nominal spacing between edges of adjacent pulses shall be 1.8 ±0.05 USEC (See 4.16.14.2.1).

3.19.12.2.2 SINGLE-PULSE REPLY POSITION. For single pulse replies, the pulse shall occupy the same timing position, ±0.05 USEC, as the third pulse of a three-pulse reply (See 4.16.14.2.2).

3.19.12.3 JAMMING REPLIES. The equipment shall have the capability of simulating Mode 4 jamming by adding from 0 to 5 front panel selected Mode 4 reply signals to a normal three-pulse Mode 4 reply (See 4.16.14.2.3).

3.19.12.3.1 TIMING. The jamming replies and the Mode 4 reply shall occur within a time period consisting of sixteen consecutive time slots spaced in 4.0 USEC intervals with the Mode 4 reply positioned in the sixteenth time slot regardless of the number of jamming replies selected (See 4.16.14.2.3.1).

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3.19.12.3.1.1 JAMMING CYCLE. The jamming pattern shown in Table I shall be repeated every fourth PRF cycle (See 4.16.14.2.3.2).

TABLE I

NO. OF SELECTED JAMMING SIGNALS	1			2			3			4			5		
TIME SLOTS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
PRF 1	X			X			X			X			X			0
PRF 2		X			X			X			X			X		0
PRF 3			X			X			X			X			X	0

X - Denotes Jamming Signal.
0 - Denotes 3 Pulse Mode 4 Reply

3.19.12.4 VARIABLE PULSE SPACING. The equipment shall provide a front panel substitute pulse select control for selecting any information pulse, the second bracket pulse of a SIF reply or the second or third pulse of a Mode 4 reply as a substitute pulse (See 4.20).

3.19.12.4.1 SPACING CONTROL. A substitute pulse spacing control shall be provided to vary the spacing of the selected substitute pulse in incremental steps as follows (See 4.16.14.4):

SPACING INCREMENTS

-0.5 ± 0.02 USEC
-0.35 ± 0.02 USEC
-0.25 ± 0.02 USEC
-0.15 ± 0.02 USEC
0.0 ± 0.02 USEC (Nominal Position)
+0.15 ± 0.02 USEC
+0.25 ± 0.02 USEC
+0.35 ± 0.02 USEC
+0.50 ± 0.02 USEC

3.19.12.4.2 CONTINUOUS SPACING. The equipment shall provide for continuously varying the spacing of the moveable reply pulses from -0.8 USEC to +0.8 USEC about the zero position (See 4.16.14.3.1).

3.19.12.5 VARIABLE PULSE WIDTH. The equipment shall provide for selecting, by a front panel control, reply video pulse widths of 0.15 USEC, 0.45 USEC, 0.50 USED and 0.90 USEC. The tolerance of each pulse width shall be ±0.05 USEC (See 4.16.14.4).

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3.19.12.5.1 CONTINUOUSLY VARIABLE WIDTH. The equipment shall be provided with a control for continuously varying the reply video pulse width from 0.15 USEC to 1.5 USEC (See 4.16.14.4.1).

3.19.12.6 REPLY GATING.

3.19.12.6.1 VARIABLE INTERNAL GATING. The equipment shall provide for selecting, by a front panel control, the number of consecutive replies passed to the front panel video output connector from 0 to 99. Provisions shall also be provided for selecting, by a front panel control, the number of consecutive replies inhibited from passing to a front panel connector from 0 to 99 (See 4.16.14.5.1).

3.19.12.6.2 EXTERNAL GATING. The equipment shall provide for external gating operation. When an external gate with the following characteristics is applied to the equipment through a front panel connector, output replies shall occur only during the gate period (See 4.16.14.5.2):

a.	Amplitude	5.0 ±2.0 volts
b.	Duration	100 to 1250 MSEC
c.	Rise time	5.0 USEC or less
d.	Decay Time	5.0 USEC or less
e.	Negative Overshoot	1.5 volts or less
f.	Quiescent DC Voltage	0 ±0.5 volts

3.19.13 MARKERS.

3.19.13.1 TIMING MARKERS. Markers, synchronized with the radar trigger (3.19.3), shall be provided at a front panel output connector for fixed-frequency operation (See 4.16.15.1).

3.19.13.1.1 MARKER CHARACTERISTICS. The timing markers shall have the following characteristics when terminated in 75 ohms ±10% (See 4.16.15.1.1):

a.	Amplitude	0.5 volts, positive, or greater
b.	Width	No more than 35% of time interval between markers
c.	Spacing	Intervals of 0.1, 1.0 and 10.0 USEC

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d. Accuracy ±0.02%

3.19.13.1.2 MARKER OUTPUT DURATION.

3.1.19.13.1.2.1 INTERNAL TRIGGER OPERATION. Marker output shall be continuous for internal trigger operation (See 4.16.15.1.2.1).

3.19.13.1.2.2 EXTERNAL TRIGGER OPERATION. Marker output shall be present for at least 500 USEC during each PRF cycle for external trigger operation (See 4.16.15.1.2.2).

3.19.13.2 SWEEP FREQUENCY MARKERS. Frequency markers, having the following characteristics when terminated in 10K ohms ±10%, shall be provided at the front panel marker output connector for use with the equipment's RF sweep frequency circuits (See 4.16.15.2):

a. Amplitude 5.0 ±1.0 volts positive

b. Width 2.0 USEC ±0.5 USEC

3.19.13.2.1 NARROW SWEEP BANDWIDTH. A marker shall be provided at center frequency, with additional markers at -5 MHz, -1 Mhz, +1 MHz, and +5 MHz about the center frequency (See 4.16.15.2.1).

3.19.13.2.2 WIDE SWEEP BANDWIDTH. Markers shall be provided the same as the narrow sweep bandwidth with additional markers at -15 MHz, -10 MHz, +10 MHz and +15 MHz about the center frequency (See 4.16.15.2.2).

3.19.13.2.3 MARKER ACCURACY. Marker accuracy shall be as follows (See 4.16.15.2.3):

MARKER	ACCURACY
Center Frequency	±0.1 MHz
+1 Mhz	±0.1 MHz
-1 MHz	±0.1 MHz
+5 MHz	±0.1 MHz
-5 MHz	±0.1 MHz
+10 MHz	±0.3 MHz
-10 MHz	±0.3 MHz
+15 MHz	±0.3 MHz
-15 MHz	±0.3 MHz

3.19.14 HIGH LEVEL RF INPUT/OUTPUT.

3.19.14.1 CONNECTORS. Two front panel type "N" female connectors, one labeled as SUM and the other labeled as DIFFERENCE, shall be provided for the equipment RF input/output channels (See 4.20).

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3.19.14.2 RF INPUT/OUTPUT CHANNEL SWITCH. The equipment shall be provided with a front panel switch to select normal or interleave operation (3.19.14.8) (See 4.20).

3.19.14.3 INPUT/OUTPUT VSWR. The VSWR of each channel shall not exceed 1.35:1 when terminated with a load impedance of 50 ohms (See 4.16.16.1).

3.19.14.4 RF OUTPUTS. The equipment shall provide fixed frequency and sweep frequency RF output signals from the SUM and DIFFERENCE input/output channels as specified herein.

3.19.14.4.1 FIXED FREQUENCY. The equipment shall provide a crystal-controlled 1090 MHz output signal with an accuracy of ±0.01 percent (See 4.16.16.2.1).

3.19.14.4.2 SWEEP FREQUENCY. The equipment shall provide a narrowband sweep output from 1085 MHz to 1095 MHz and a wideband sweep output from 1075 MHz to 1105 MHz (See 4.16.16.2.2).

3.19.14.4.3 SWEEP BLANKING. For RF sweep operation, modulation of the equipment RF output shall only occur during the time the RF sweep changes frequency (See 4.16.16.2.3).

3.19.14.5 CHANNEL ATTENUATION CONTROLS. Two separate front panel attenuators, each calibrated in dB below 1 milliwatt under matched load conditions, shall be provided for continuously varying the SUM and DIFFERENCE RF output levels between 0 and 90 dB below 1 milliwatt when matched to a 50 ohm impedance load (See 4.16.16.3).

3.19.14.5.1 ATTENUATOR SCALES. Each attenuator scale shall contain 90 divisions at one dB intervals and be labeled at each tenth division, i.e., 0, -10, -20, -30, -40, -50, -60, -70, -80, and -90 (See 4.20).

3.19.14.5.2 DIFFERENTIAL ATTENUATION. The differential attenuation between any two consecutive 10 dB divisions shall be within 10.0 ±0.5 dB (See 4.16.16.3.1).

3.19.14.6 FIXED FREQUENCY RF OUTPUT LEVEL ACCURACY. The equipment output level for fixed frequency operation and nominal pulse width shall be accurate within ±1.0 dB of the attenuator setting (See 4.16.16.4).

3.19.14.7 SWEEP FREQUENCY RF OUTPUT LEVEL FLATNESS. The equipment output level for sweep frequency operation shall be flat within 1.0 dB peak-to-peak over the sweep bandwidths (See 4.16.16.5).

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3.19.14.7.1 SWEEP OFFSET. For the same attenuator setting as for fixed frequency operation (3.19.14.6) the output level of the RF sweep at 1090 MHz center frequency shall be within ±0.5 dB of the fixed frequency level (See 4.16.16.5.1).

3.19.14.8 INTERLEAVE RF OUTPUT. When interleave operation is selected by a front panel control, the SUM channel RF output and DIFFERENCE channel RF output shall be internally combined and provided at the SUM channel RF input/output connector (3.19.14.1) (See 4.16.16.6).

3.19.14.8.1 SUM LEVEL. The SUM attenuator shall control the level of the combined RF output signal from 0 to -90 dBm (See 4.16.16.6.1).

3.19.14.8.2 DIFFERENCE LEVEL. The DIFFERENCE attenuator shall control the level of the DIFFERENCE output signal from a level equal to the SUM channel output signal ±0.5 dB to at least 90 dB below the level of the SUM channel output signal. The minimum output signal level due to any combination of attenuator settings is not required to be below -90 dBm (See 4.16.16.6.2).

3.19.14.8.3 INTERLEAVE LEVEL SET ACCURACY. For interleave operation the DIFFERENCE channel output shall be within ±1.0 dB of the SUM channel output when both of the attenuators are set for output levels of 0 dBm (See 4.16.16.6.3).

3.19.14.9 MODULATION.

3.19.14.9.1 MODULATION SELECT SWITCH. A multi-position front panel select switch shall be provided to select the internal modulating signals for the equipment's RF outputs. An OFF position shall be included (See 4.20).

3.19.14.9.1.1 INTERNAL REPLY MODULATION SIGNALS. The equipment's reply modulation shall include the video signals specified in 3.19.12 through 3.19.12.5.1 (See 4.16.16.7.1).

3.19.14.9.1.2 CONTINUOUS WAVES (CW). Continuous Wave (CW) RF signals shall be internally generated and provided as SUM and DIFFERENCE channel outputs (See 4.16.16.7.2).

3.19.14.9.1.3 HIGH-RATE MODULATION SIGNAL. A 30 KHz ±10% signal shall be internally generated and provide modulation for the equipment's SUM and DIFFERENCE RF outputs (See 4.16.16.7.3).

3.19.14.9.2 EXTERNAL MODULATION. A front panel connector shall be provided to accept external modulation signals for the SUM channel, having characteristics as specified in 3.19.12.1 and 3.19.12.2.

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External modulation operation shall be possible whether or not internal pulse modulation has been selected (See 4.16.16.7.4).

3.19.14.9.3 INTERLEAVED MODULATION. The modulated RF output from the DIFFERENCE channel shall occur 0.70 ±0.05 USEC after the SUM channel modulation for equipment interleave operation (3.19.14.8) (See 4.16.16.7.5).

3.19.14.9.4 MODULATION FIDELITY. The equipment's RF output signals shall have the following characteristics when modulated by the video signals specified in 3.19.12 through 3.19.12.6.2 and 3.19.14.9.3 (See 4.16.16.7.6):

a.	Rise Time	0.05 to 0.1 USEC
b.	Decay Time	0.05 to 0.2 USEC
c.	Width	As specified in 3.19.12.1 and 3.19.12.2

3.19.14.9.5 MODULATOR DELAY. The delay between the external modulation pulse applied to the equipment input (3.19.14.9.2) and the resulting modulated RF output shall not exceed 0.3 USEC (See 4.16.16.7.7).

3.19.14.9.6 CW LEAKAGE. When pulse modulation is selected. the equipment's CW leakage from either RF input/output channel shall be no greater than -90 dBm (See 4.16.16.7.8).

3.19.14.9.7 MODULATOR ON-OFF RATIO. The equipment modulators shall have on-to-off ratios of 60 dB or greater (See 4.16.16.7.9).

3.19.15 LOW LEVEL RF INPUT. The equipment shall be provided with a low level RF input channel whose input connector is of a different type than the high level input (3.19.14) to minimize the possibility of connecting a high level signal to this input (See 4.20).

3.19.16 MEASUREMENT CAPABILITY. The equipment when used with the Oscilloscope AN/USM-281 or equivalent shall be capable of measuring pulse fidelity, RF power, RF frequency and PRF characteristics of IFF interrogation signals (See 4.16.17).

3.19.16.1 DEMODULATOR SENSITIVITY.

3.19.16.1.1 HIGH LEVEL. The equipment shall demodulate an RF input of +18 dBw to +33 dBw and shall produce at the front panel a

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demodulated video output pulse of not less than 1 volt amplitude into a 75 ohm ±5% impedance (See 4.16.17.1.1).

3.19.16.1.2 LOW LEVEL. The equipment shall demodulate an RF input of -12 dBm to +3dBm and shall produce a front panel demodulated video output pulse of not less than 1 volt amplitude into a 75 ohm ±5% impedance (See 4.16.17.1.2).

3.19.16.1.2.1 EXTENDED LOW LEVEL RANGE. The low power input capabilities shall be extended to cover the range of +3 dBm to +48dBm using the attenuators specified in 3.20.5 (See 4.16.17.3).

3.19.16.2 PULSE FIDELITY. The distortion in the demodulated video output's pulse rise time, fall time and width shall not deviate from a corresponding output of a standard biased detector by more than ±0.02 USEC or ±5% whichever tolerance is greater over the frequency range of 1025 MHz to 1035 MHz (See 4.16.17.2).

3.19.16.3 PULSE POWER MEASUREMENT CAPABILITY. The equipment shall have the following pulse power measurement capability with the readout indicated on the equipment meter specified in 3.19.16.6 for the high level inputs (3.19.16.1.1) and with a conversion chart (3.20.7) and a meter indication for the low level inputs (3.19.16.1.2 and 3.19.16.1.2.1) (See 4.16.17.3):

POWER LEVEL INPUTS

	MEASUREMENT TOLERANCE	MINIMUM	MAXIMUM
Power Range	---	-12 dBm	63 dBm
Frequency Range	---	1025 MHz	1035 MHz
Accuracy High Level	±1.0 dB	18.0 dBw	33.0 dBw
Accuracy Low Level	±1.0 dB	-12 dBm	3 dBm
Accuracy Extended Low Level	±1.0 dB*	3.1 dBm	47.9 dBm

*Plus accuracy of the accessory attenuators.

3.19.16.4 FREQUENCY MEASUREMENT CAPABILITY. The equipment, using the Oscilloscope (3.19.16) as a readout indicator, shall have the following frequency measurement capability for the input power levels specified in 3.19.16.1.1 through 3.19.16.1.2.1 (See 4.16.17.4):

FREQUENCY INPUTS

	MEASUREMENT TOLERANCE	MINIMUM	MAXIMUM
Frequency Range	---	1015 MHz	1045 MHz
Accuracy	±0.6 MHz	1015.0 MHz	1024.9 MHz
Accuracy	±0.2 MHz	1025.0 MHz	1035.0 MHz
Accuracy	±0.6 MHz	1035.1 MHz	1045.0 MHz

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3.19.16.5 PRF MEASUREMENT. The equipment shall be capable of measuring repetition rates of external video signals (3.19.16.8), internally generated reply signals (3.19.1.1) or demodulated challenges (3.19.1.3). Measurement accuracy shall be ±5% or ±5 degrees, whichever tolerance is greater from 1/10th full scale to full scale (See 4.16.17.5).

3.19.16.6 METER. The equipment shall contain a multi-scale meter with scales marked in dBw and Hz to indicate challenge/reply PRF's and power levels (See 4.16.17.4 and 4.16.17.5).

3.19.16.7 SELF-TEST.

3.19.16.7.1 SIGNAL ROUTING. The internally generated RF signals shall be routed from either RF input/output channel connector (SUM, DIFFERENCE) to the low power input connector through an RF jumper cable specified in 3.20.3 (See 4.16.17.6.1).

3.19.16.7.1.1 PULSE FIDELITY MEASUREMENT. The pulse rise time, fall time and width characteristics of 3.19.14.9.4 shall not be degraded by more than ±0.02 USEC or ±5% whichever tolerance is greater, when the equipment's RF outputs are demodulated during self-test operation (See 4.16.17.6.1.1).

3.19.16.7.1.2 POWER MEASUREMENT ACCURACY. The measurement accuracy of the equipment peak power output at the -5 dBm level shall be ±0.5 dB (See 4.16.17.6.1.2).

3.19.16.7.2 BUILT-IN-TEST (BIT) FOR FREQUENCY MEASUREMENT. The equipment shall have built-in capabilities to measure the accuracy of its frequency measurement circuitry to a tolerance of ±0.3 MHz (See 4.16.17.6.2).

3.19.16.8 EXTERNAL VIDEO IN. A front panel connector, with an input impedance of 75 ohms ±10% or 2000 ohms ±10% as selected by a front panel switch, shall be provided and shall accept external video signals having the following characteristics (See 4.16.17.7):

a.	Polarity	Positive or Negative
b.	Amplitude	3V Minimum 20V Maximum
c.	Repetition Rate	Between 101 Hz and 2000 Hz
d.	Pulse Groups	One or more pulses within a 100 USEC period
e.	Individual Pulse Width	0.25 USEC Minimum, 15 USEC Maximum

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f.	Individual Pulse Rise Time	0.1 USEC Maximum
g.	Individual Pulse Decay Time	1.0 USEC Maximum

3.19.17 POWER SUPPLY FAULT INDICATOR. The equipment shall have a front panel indicator which will illuminate when any of the DC working voltages developed by the equipments power supply deviate more than ±1.0 volt from nominal. The indicator shall not illuminate for variations within ±0.25 volts of nominal. For variations between ±0.25 volts and ±1.0 volts of nominal, the indicator may or may not illuminate (See 4.16.18).

3.19.18 POWER SUPPLY SHORT CIRCUIT PROTECTION. The equipment shall have the capability to prevent power supply damage when any of the DC working voltages become shorted (See 4.16.19).

3.20 ACCESSORIES. Each equipment shall contain the accessories specified in 3.20.1 through 3.20.8.

3.20.1 RF CABLE. Two 8-foot ±0.25 ft. lengths of RF cable, type RG-212/U having a UG-18/U, or equivalent, plug on each end, shall be provided. This cable shall be marked to indicate its attenuation at 1030 MHz and 1090 MHz (See 4.17.1).

3.20.2 VIDEO CABLES. Four 5-foot ±0.25 ft. lengths of RF cables, type RG-62A/U, each having a UG-260F/U plug on each end, shall be provided (See 4.17.1).

3.20.3 RF JUMPER CABLE. One 12-inch ±1 inch RF cable RG-223/U, having suitable connectors to connect the SUM or DIFFERENCE channels to the equipment low power input/output for self-test purposes, shall be provided (See 4.17.1).

3.20.4 PRIMARY POWER CABLE. One 8-foot ±0.25 ft. length of AC power cable with suitable connectors shall be provided (See 4.17.1).

3.20.5 FIXED ATTENUATORS. A 15.0 ±0.2 dB attenuator and a 30.0 ±0.3 dB attenuator, each having a connector to mate with the low power input connector (3.19.15) on one end and a connector functionally and electrically equivalent to the low power input connector on the other end shall be provided with the following characteristics (See 4.16.17.3 and 4.17.3).

a.	Frequency Range	1010 MHz to 1050 MHz.
b.	Impedance	50 ±3 ohms.

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c. VSWR (Maximum) 1.2:1.

d. Power Dissipation Rating 1.0 watt or greater.

3.20.6 VIDEO TERMINATIONS. Three feed through-type terminations, each having a connector on one end to mate with a UG-912 and a connector on the other end to mate with a UG-260, shall be provided as follows: (See 4.17.2):

QUANTITY	DC RESISTANCE
2	75 ohms ±5%
1	93 ohms ±5%

3.20.7 CONVERSION CHART. A chart shall be provided to convert from the meter indication to the actual input power level applied to the low power input connector (See 4.16.17.3).

3.20.8 VIDEO PROBE. One video probe, Model 2739-C-48, Pomona Electronics Co. Inc., or equivalent, shall be provided (See 4.20).

3.21 BURN-IN/DEBUGGING. Each unit on order shall be capable of meeting the requirements herein, after subjection to the burn-in/debugging as specified (See 4.5.1).

3.22 OPERABILITY. The unit(s) shall be operable as specified, after applicable processes and inspections are completed (See 4.18).

3.23 INTERCHANGEABILITY. Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such items or of the units (See 4.19).

3.24 WORKMANSHIP. The equipment shall be manufactured and assembled in accordance with the following requirements and paragraphs herein (See 4.20):

	MIL-STD-454
Cable, Coaxial(RF)	Requirement 65
Cable Multiconductor	Requirement 66
Controls	Requirement 28
Fungus-Inert Materials	Requirement 4
Fuses and Fuse Holders	Requirement 39
Electrical Overload Protection	Requirement 8
Soldering	Requirement 5
Workmanship	Requirement 9

4. QUALITY ASSURANCE PROVISIONS.

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4.1 RESPONSIBILITY FOR INSPECTION. Unless Otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF INSPECTION. Inspection shall be classified as follows:

a. First Article Inspection. Does not include packaging (See 4.3).

b. Inspection Covered by Subsidiary Documents. (See 4.4).

c. Quality Conformance Inspection.

(1) Quality conformance inspection of equipment before packing (See 4.5).

(2) Quality conformance inspection of packing (See 4.21).

4.3 FIRST ARTICLE. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 FIRST ARTICLE UNITS. The contractor shall furnish first article units of the complete Test Set, Radar Set AN/TPM-25A as specified in the contract.

4.3.2 FIRST ARTICLE INSPECTION. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, burn-in/debugging and reliability qualification specified in 4.5, and the inspections specified for Group A, Group B, and Group C (See Tables, I, II, III, respectively). The inspection shall be performed in the following order: 4.4, burn-in/debugging, Group A and Group B for all units, and designated units for specific subgroups within Group C, in accordance with 4.5.4.1 The reliability qualification may be conducted concurrent with Group C inspection.

4.3.3 FIRST ARTICLE DATA. The first article test plan and test report(s) shall be as required in the contract.

4.4 INSPECTION COVERED BY SUBSIDIARY DOCUMENTS. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

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INSPECTIONS	REQT PARA
General Requirements	3.3
Parts, Materials and Processes	3.4
Finish, Protective	3.5
Marking	3.6

4.5 QUALITY CONFORMANCE INSPECTION OF EQUIPMENT BEFORE PACKAGING. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 BURN-IN/DEBUGGING. All units presented to the Government for acceptance shall undergo a 48 hour burn-in. The burn-in shall be conducted under the same conditions as the reliability qualification test (4.5.5). The last half of the burn-in period shall be failure free. If a failure occurs during the last half of the burn-in period, the equipment shall be repaired, the repair shall be verified, and the equipment shall be returned to burn-in. This procedure shall be followed until the 24 hour failure free period is achieved. All failures and maintenance actions shall be recorded by the contractor and the records shall be supplied to the Government as specified in the contract.

4.5.2 GROUP A INSPECTION. Each unit on contract shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQL's indicated in Table I.

4.5.2.1 ORDER OF INSPECTION WITHIN GROUP A. Group A inspection shall be performed in an order satisfactory to the Government.

4.5.3 GROUP B INSPECTION. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table II and to the special inspection levels of Table I of MIL-STD-105.

4.5.3.1 GROUP B SAMPLING PLANS. The Group B AQL shall be 6.5% and the inspection level shall be S4.

4.5.3.2 ORDER OF INSPECTION WITHIN GROUP B. Group B inspection shall be performed in an order satisfactory to the Government.

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4.5.4 GROUP C INSPECTION. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table III. No defects shall be allowed.

TABLE I - GROUP A INSPECTION

INSPECTION	REQUIREMENT PARAGRAPH	TEST PARAGRAPH	AQL MAJOR	AQL MINOR
Visual and Mechanical	3.24	4.20	1%	4%
Printed Circuit Board Assemblies			1%	
TS-3394/TPM-25A			1%	
Attenuators			1%	
Video Probe			1%	
Calibration Chart			1%	
Case, Instrument			1%	
Cable Assemblies			1%	
AN/TPM-25A			1%	
Electrical				
Internal Triggered PRF	3.19.1.1 thru 3.19.1.1.1	4.16.1 thru 4.16.1.1		
Decode Triggered PRF	3.19.1.3 thru 3.19.1.3.3	4.16.3 thru 4.16.3.3		
Sweep Frequency Mode PRF	3.19.1.4 thru 3.19.1.4.1	4.16.4.1 thru 4.16.4.1		
Radar Trigger Characteristics	3.19.3	4.16.6		
Selective Identification Feature (SIF)	3.19.4	4.16.7		
Mode 4 Gain Time Control (GTC) Trigger	3.19.5	4.16.8		
Scope Triggers	3.19.6	4.16.9		
Fixed Frequency Scope Trigger	3.19.6.1	4.16.9.1		
Sweep Sync Scope Trigger	3.19.6.2	4.16.9.2	4.0% for the Combined Group	
Decode Scope Trigger	3.19.6.3	4.16.9.3		
Delayed RF Sweep Trigger	3.19.7	4.16.9.4		
System Timing	3.19.8 thru 3.19.8.2.1	4.16.10 thru 4.16.10.2.1		
Decode Operation	3.19.8.2.2	4.16.10.2.2		
Triggered Operation	3.19.8.3.1	4.16.10.3.1		
Reply Position	3.19.8.3.1.3	4.16.10.3.1.3		
Decode Operation	3.19.8.3.2	4.16.10.3.2		
Timing Interlock	3.19.9	4.16.11		

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Indicator	3.19.9.1	4.16.11.1	
Mode 4 Challenge Video Output	3.19.10	4.16.12	
Pulse Train Timing	3.19.10.1	4.16.21.1	(Continued from Previous Page)
Pulse Characteristics	3.19.10.3	4.16.12.3	
SIF Challenge Video Output	3.19.11	4.16.13	
Challenge Video Pulse Spacing	3.19.11.2	4.16.13.1	
Challenge Video Timing	3.19.11.3	4.16.13.2	
Video Characteristics	3.19.12.1 thru 3.19.12.5	4.16.14.1 thru 4.16.14.5	
Mode 4	3.19.12.2 thru 3.19.12.3.1.1	4.16.14.2 thru 4.16.14.2.3.2	
Variable Pulse	3.19.12.4.1 thru 3.19.12.5.1	4.16.14.3 thru 4.16.14.4.1	
Variable Internal Gating	3.19.12.6.1	4.16.14.5.1	
External Gating	3.19.12.6.2	4.16.14.5.2	
Markers	3.19.13.1 thru 3.19.13.2.3	4.16.15.1 thru 4.16.15.2.3	
RF Outputs	3.19.14.4.1 thru 3.19.14.4.2	4.16.16.2.1 thru 4.16.16.2.2.2	
Channel Attenuation Controls	3.19.14.5	4.16.16.3	
RF Output	3.19.14.6 thru 3.19.14.8.3	4.16.16.4 thru 4.16.16.6.3	
Modulation	3.19.14.9.1.1 thru 3.19.14.9.4	4.16.16.7.1 thru 4.16.16.7.6	
Measurement Capability	3.19.16.1.1 thru 3.19.16.5	4.16.17.1.1 thru 4.16.17.5	
Self-Test	3.19.16.7.1 thru 3.19.16.7.2	4.16.17.6.1 thru 4.16.17.6.2	
External Video In	3.19.16.8	4.16.17.7	
Cable Assemblies	3.20.1 thru 3.20.4	4.17.1	1%
Power Switch	3.8.1	4.7.1	1%
Power Indicator	3.8.3	4.7.2	1%
Operability (Operational Inspection)	3.22	4.18	1%

*All electrical defects considered major.

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TABLE II - GROUP B INSPECTION

INSPECTION	REQUIREMENT PARAGRAPH	TEST PARAGRAPH	AQL MAJOR	AQL MINOR
Pulse Repetition Frequency (PRF)	3.19.1.1.2 thru 3.19.1.2.1	4.16.1.2 thru 4.16.2.1		
Radar Trigger	3.19.2	4.16.5		
Adjustment	3.19.4.1	4.16.7.1		
Jitter	3.19.6.1.1	4.16.9.1.1		
Reply Timing Adjustment	3.19.8.2.1.1	4.16.10.2.1.1		
Reference Trigger Timing Adjustment	3.19.8.3.1.1	4.16.10.3.1.1		
Timing Adjustment Synchronization	3.19.8.3.1.2	4.16.10.3.1.2		
Jitter	3.19.10.2	4.16.12.2		
Garble Variable Pulse Spacing	3.19.12.1.5	4.16.14.1.6.1		
Sweep Blanking	3.19.14.4.3	4.16.16.2.3		
Differential Attenuation	3.19.14.5.2	4.16.16.3.1	6.5% for the Combined Group	
Sweep Offset	3.19.14.7.1	4.16.16.5.1		
Modulation	3.19.14.9.5 thru 3.19.14.9.7	4.16.16.7.7 thru 4.16.16.7.9		
Power Supply Fault Indicator	3.19.17	4.16.18	6.5% for the Combined Group	
Power Supply Short Circuit Protection	3.19.18	4.16.19		
Video Terminations	3.20.6	4.17.2	6.5%	
Input Power Source	3.8	4.7	6.5%	
Interchangeability	3.23	4.19	6.5%	

4.5.4.1 SAMPLING FOR GROUP C INSPECTION. Sample units shall be selected at random for each group of inspections in Table III. The first samples shall be selected at the start of the contract from the first quality conformance inspection production lot(s).

4.5.4.1.1 SUB-GROUP I. For this subgroup, two units from the first (10) production units shall be selected for the first Group C inspection. For subsequent inspections two (2) units from each fifty (50) production units shall be selected.

4.5.4.1.2 SUB-GROUP II. For this subgroup, one (1) unit from the first ten (10) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, one (1) unit shall be selected when one-half of the units on order have been accepted by the Government.

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4.5.4.1.3 SUB-GROUP III. Same as Sub-Group II.

4.5.4.1.4 SUB-GROUP IV. Same as Sub-Group II.

4.5.4.1.5 SUB GROUP V. Same as Sub-Group II.

4.5.4.1.6 SUB-GROUP VI. For this subgroup, two (2) units shall be selected from the first ten (10) production units. For subsequent Group C Inspection, two (2) units shall be selected when one half of the units on order have been accepted by the Government.

4.5.4.2 ORDER OF INSPECTION WITHIN GROUP C. Group C inspection shall be performed in the order shown in each sub-group in Table III for Sub-Group II thru Sub-Group VI. Sub-Group I inspection shall be performed in the order satisfactory to the Government.

4.5.4.3 GROUP C FAILURES. Actions required relative to Group C failures shall be as specified in the contract.

4.5.4.4 REINSPECTION OF CONFORMING GROUP C SAMPLE UNITS. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

TABLE III - GROUP C INSPECTION

INSPECTION	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Sub-Group I		
Warm-Up Time	3.17	4.14
Input/Output VSWR	3.19.14.3	4.16.16.1
Front Panel Protection	3.7	4.6
Extender Card	3.9	4.8
Weight	3.11	4.10
Volume	3.12	4.11
Compatibility	3.14 thru 3.14.3	4.12
Fixed Attenuators	3.20.5	4.17.3
Sub-Group II		
High Temperature	3.18.1.1	4.15.1.1
Low Temperature	3.18.1.2	4.15.1.2
Altitude	3.18.2	4.15.2
Humidity	3.18.3	4.15.3
Sub-Group III		
Vibration	3.18.4	4.15.4
Shock	3.18.5	4.15.5

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SUB-GROUP IV Salt Fog	3.18.7	4.15.7
SUB-GROUP V Fungus	3.18.6	4.15.6
SUB-GROUP VI EMI	3.16	4.13

NOTE: Units selected for Sub-Group I inspection may be used for subsequent sub-group testing.

4.5.5 RELIABILITY QUALIFICATION TESTING. The contractor shall conduct a reliability qualification test in accordance with Test Plan V, Test Level B of MIL-STD-781, using a specified MTBF of 900 hours. The test shall be performed on three units (See 3.15).

4.5.5.1 REJECT DECISION. In the event of a reject decision at any time during the test, the paragraph entitled Reject in Reliability Qualification (Demonstration) Phase of MIL-STD-781, shall be followed. The test shall be discontinued, the Government shall be notified, corrective action shall be taken, and a retest shall be done.

4.5.5.2 FAILURE DEFINITION. A failure is defined as any inability of the equipment to perform its required function, including any departure from performance as required in this specification.

4.5.5.3 PERFORMANCE MEASUREMENTS. The following performance parameters shall be measured at least once daily on each equipment during reliability testing and burn-in:

PARAMETER	REQUIRED PARAGRAPH
Radar Trigger Characteristics	3.19.3
SIF Triggered Operation	3.19.8.2.1
Mode 4 Three Pulse Reply Spacing	3.19.12.2.1
Marker Output Internal Trigger Operation	3.19.13.1.2.1
Wide Sweep Bandwidth	3.19.13.2.2
RF Fixed Frequency Accuracy	3.19.14.4.1
RF Sweep Frequency Output	3.19.14.4.2
Self-Test Power Measure Accuracy	3.19.16.7.1.2

4.6 FRONT PANEL PROTECTION. The equipment shall be oriented such that it rests on the front panel handles. Verify that no control, indicator or electrical connector comes in contact with the resting surface (See 3.7).

4.7 INPUT POWER SOURCE: Power shall be applied to the equipment and the tests 4.16.17.6.1.1, 4.16.17.6.1.2, and 4.16.17.6.2 performed at the following combinations of voltages and frequencies (See 3.8):

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Frequency Hz (±1%)	Volts VAC (±%) 103.5 VAC	115 VAC	126.5 VAC
45 Hz	X	X	X
60 Hz		X	
66 Hz		X	
380 Hz		X	
420 Hz	X	X	X

4.7.1 POWER SWITCH. Verify that the equipment is energized only when switch is in the "ON" position (See 3.8.1).

4.7.2 POWER INDICATOR. Verify the energizing of power indicator lamp with the application of the AC power source (See 3.8.3).

4.8 EXTENDER CARD. Substitute the extender card for one of the equipment's printed circuit cards. Insert the replaced printed circuit card into the mating connector of the extender card and perform the test of 4.16.17.6.2. Verify that all components mounted on the printed circuit card are accessible when the extender card is used. Further verify that keying or other mechanical restraints are provided to prevent improper insertion of the extender card into the replaced printed circuit card position and to prevent improper insertion of the printed card into the extender card's mating connector. Repeat the above for each of the equipment's printed circuit cards (See 3.9).

4.9 STANDARD TEST CONDITION. Unless otherwise specified, inspections shall be performed at prevailing ambient temperature, atmospheric pressure and relative humidity, with the following equipment control settings:

Input Power	115 VAC ±10%, 60 Hz ±5%
Reply Video	SIF
Reply Pulse Width	0.45 USEC
Mode of Operation	Fixed Frequency - Normal
RF Output	SUM 0 dBm
	DIFFERENCE 0 dBm
PRF	400 Hz

4.10 WEIGHT. Verify the weight of the equipment (See 3.11).

4.11 VOLUME. Verify the volume of the equipment (See 3.12).

4.12 EQUIPMENT COMPATIBILITY. Connect the equipment front panel ground terminal to the oscilloscope ground terminal. Verify that the equipment chassis is grounded to the oscilloscope chassis. Interconnect the AN/TPM-25A and AN/TPM-24(V) with the AN/TPX-46 and measure the equipment characteristics specified below; then substitute the AN/TPX-50 for the AN/TPX-46 and again measure these characteristics. After completion of these measurements, interconnect the AN/TPM-25A with the AN/APM-123 and measure the equipment characteristics specified below.

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a. AN/TPX-46 and AN/TPX-50 Interrogators
Transmitter Power for both SUM and DIFFERENCE channels.
Mode 4 Pulse Train Droop.
Transmitter Frequency.
Pulse Shape and Spacing (SIF).
SUM and DIFFERENCE Receiver Sensitivity.
Receiver Interleave Pulse Train Fidelity.
Receiver RSLS Comparison and Crossover.
Receiver Gate Duration.
Receiver GTC Operation.
Video Target Widths and Amplitudes.
Pulse Width Discrimination.
PRF Countdown Operation.
SIF Decoder Operation and Characteristics.
Degarbling Characteristics.
Mode 4 Decoder Operation and Characteristics.
Target Threshold Level.
Friend Threshold Level.
Threshold with Mode 4 Jamming.

b. AN/APM-123 Test Set
Transmitter Power.
Transmitter Frequency.
Pulse Shape and Spacing.
ISLS Characteristics.
Receiver Sensitivity.
Receiver Bandpass Characteristics.
Decoder Characteristics.
Reply Evaluator Thresholds.

4.13 ELECTROMAGNETIC INTERFERENCE TESTS. The equipment shall be tested for compliance with the EMI requirements for Section 3. The test methods and procedures used shall be in accordance with MIL-STD-462, Notice 3, as implemented by a contractor prepared, Government approved, EMI test plan as specified in the contract (See 3.16).

4.14 WARM-UP. Apply power to the equipment and within ten minutes conduct the test of 4.16.17.6.1.2 (See 3.17).

4.15 SERVICE CONDITIONS.

4.15.1 TEMPERATURE.

4.15.1.1 HIGH TEMPERATURE. The equipment shall be subjected to a high temperature test in accordance with MIL-STD-810, Method 501, Procedure I. Except for the requirement of 3.19.14.7, the performance of the equipment during the high temperature test shall be as specified in Table IV and shall be demonstrated at 55°C with an

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input power source of 126.5 VAC, 420 Hz. During this test, the performance tolerances for requirement 3.19.14.7 shall be ±3 dB. The performance of the equipment, before and after high temperature test, shall be as specified in Table IV and shall be demonstrated under standard Test conditions as specified in 4.9 (See 3.18.1.1).

4.15.1.2 LOW TEMPERATURE. The equipment shall be subjected to a low temperature test in accordance with MIL-STD-810, Method 502, Procedure I. Except for the requirement of 3.19.14.7, the performance of the equipment during the low temperature test shall be as specified in Table IV and shall be demonstrated at 0°C with an input power source of 103.5 VAC, 45 Hz. During this test, the performance tolerances for requirement 3.19.14.7 shall be ±3 dB. The performance of the equipment, before and after the low temperature test, shall be as specified in Table IV and demonstrated under standard test conditions specified in 4.9 (See 3.18.1.2).

4.15.2 ALTITUDE. The equipment shall be subjected to a low pressure test in accordance with MIL-STD-810, Method 500, Procedure I. The performance of the equipment during the low pressure test shall be as specified in Table IV and shall be demonstrated at 10,000 feet equivalent pressure. The performance of the equipment, before and after the low pressure test, shall be as specified in Table IV and shall be demonstrated under standard test conditions as specified in 4.9 (See 3.18.2).

4.15.3 HUMIDITY. The equipment shall be subjected to a humidity test in accordance with Method 507, Procedure II of MIL-STD-810. Monitoring tests, for the purpose of determining continued equipment operation, shall be permitted during the last 5-hour period of each humidity cycle in accordance with the instructions specified in Step 6 of Procedure II. The performance during the humidity test shall be determined at Step 8. Except for the requirement of 3.19.14.7, the equipment's performance shall be determined in accordance withthe test specified in Table IV. The performance tolerances for requirement 3.19.14.7, shall be 3.19.10.6.2 shall be ±3 dB. After the completion of the fifth cycle, the equipment shall be conditioned in accordance with Step 2 and tested as specified in Table IV under standard test conditions. Full performance shall be required (See 3.18.3).

4.15.4 VIBRATION. Prior to initiating the vibration test, the equipment shall be tested in accordance with Table IV. The equipment, while non-operating, shall then be subjected to a simple harmonic motion having an amplitude of 0.015 inches (0.030 inches maximum total excursion), between the limits of 5 to 55 Hz. The rate of frequency change shall be such that a complete cycle (5/55/5) will consume 8 minutes. The test item shall be vibrated for 15 cycles(2 hours) along each of the 3 major axes of the equipment for a total

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test time of 6 hours. At the completion of the test, the equipment shall be visually inspected for mechanical failure. The equipment shall then be tested in accordance with the performance tests of Table IV under standard test conditions (See 3.18.4).

4.15.5 SHOCK. (See 3.18.5).

4.15.5.1 BENCH HANDLING. The equipment shall be tested in accordance with MIL-STD-810, Method 516.1, Procedure V. The performance tests before and after shall be conducted as specified in Table IV (See 3.18.5.1).

4.15.5.2 BASIC DESIGN. The equipment, with cover attached, shall be tested in accordance with MIL-STD-810, Method 516.1, Procedure 1. The shock pulse characteristics shall be in accordance with Figure 516.1-2, except that the amplitude shall be 15 g's and the time duration 11 milliseconds. The performance tests before and after shall be conducted as specified in Table IV (See 3.18.5.2).

4.15.6 FUNGUS. The unit, with its cover separated from the equipment, shall be tested in accordance with Method 508, Procedure 1 of MIL-STD-810. No cleaning of the unit(s) is permitted for 72 hours prior to the fungus test. Handling prior to and during testing shall be accomplished without contamination of the equipment. The interior of the case shall also be sprayed and the unit(s) shall be subjected to the test in an open condition. There shall be abundant growth colonization on 50 percent or more of the area of the control items after 14 and 28 days of test. At the conclusion of the incubation period, the unit(s) shall be inspected and show no more than 0.015 square inch in area, of sparse microbial growth as evidenced by growth colonization (which includes branching and sporulation) on or within each cubic foot or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination shall not be included in this evaluation (See 3.18.6 and 6.5).

4.15.7 SALT FOG. The unit, with its cover separated from the equipment, shall be subjected to the salt fog test in accordance with Method 509 of MIL-STD-810 (See 3.18.7).

4.16 ELECTRICAL PERFORMANCE TESTS. (See 3.19).

4.16.1 INTERNAL TRIGGERED PRF. While varying the Pulse Repetition Time(PRT) front panel controls from minimum to maximum setting, maesure the period of the equipment's PRF cycle (See 3.19.1.1).

4.16.1.1 ACCURACY. While performing the test of 4.16.1, verify the accuracy of the indicated PRT settings (See 3.19.1.1.1).

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TABLE IV

	Low Temp		High Temp		Alt.		Humi-dity		Vib.		Shock Bench		Shock Basic	
Req. Para.	Before/After	During	Before/After	During	Before/After	During	Before/After	During	Before/After	During	Before/After	During	Before/After	During
3.19.1.1	X	X	X	X			X	X	X		X		X	
3.19.1.1.1	X	X	X	X			X	X	X		X		X	
3.19.1.4	X	X	X	X			X	X	X				X	
3.19.2					X	X					X			
3.19.3	X	X	X	X			X	X	X				X	
3.19.5	X	X	X	X										
3.19.8.1	X	X	X	X					X					
													X	
3.19.8.2.1	X	X	X	X	X	X	X	X	X				X	
3.19.8.2.2	X	X	X	X			X	X	X				X	
3.19.8.3.1.3	X	X	X	X			X	X	X				X	
3.19.10	X	X	X	X			X	X	X				X	
3.19.11.2	X	X	X	X										
3.19.12.1.4	X	X	X	X	X	X	X	X	X		X		X	
3.19.12.2.1	X	X	X	X			X	X	X				X	
3.19.12.6.2	X	X	X	X			X	X						
3.19.13.1	X	X	X	X			X	X	X				X	
3.19.13.1.2.1	X	X	X	X			X	X	X		X		X	
3.19.13.2	X	X	X	X			X	X	X				X	
3.19.13.2.1	X	X	X	X			X	X	X				X	
3.19.13.2.2	X	X	X	X			X	X	X				X	
3.19.13.2.3	X	X	X	X			X	X	X				X	
3.19.14.4.1	X	X	X	X			X	X	X		X		X	
3.19.14.6	X	X	X	X			X	X	X		X		X	
3.19.14.7	X	X	X	X			X	X	X				X	
3.19.14.9.4	X	X	X	X			X	X						
3.19.16.2	X	X	X	X			X	X						
3.19.16.3	X	X	X	X			X	X	X				X	
3.19.16.4	X	X	X	X			X	X						
3.19.16.7.1.2					X	X			X		X		X	
3.19.16.7.2									X		X		X	

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4.16.1.2 JITTER. Measure the period-to-period jitter of the equipment's PRF cycle (See 3.19.1.1.2).

4.16.2 EXTERNAL TRIGGERED PRF. Set the PRF select switch (3.19.1) for external trigger operation. Using a pulse generator, apply an external trigger having characteristics as specified in 3.19.1.2. Verify that the operating PRF of the equipment is the same as the PRF of the external trigger within the tolerance of 3.19.16.5 (See 3.19.1.2).

4.16.2.1 EXTERNAL PRF GENERATION. While conducting the test of 4.16.2, verify that only one equipment PRF cycle is generated for each external trigger (See 3.19.1.2.1).

4.16.3 DECODE TRIGGERED PRF. Set the PRF select switch (3.19.1) for decode trigger operation and the decode enable switch (3.19.1.3.3) to ON. Apply MOdes 1, 2, 3/A, 4 and C RF challenges, respectively, to the low power input (3.19.15). Vary the mode selection switch 3.19.1.3.2) to correspond with the mode of the applied challenge signal. Verify that the operating PRF of the equipment is the same as the PRF of the applied challenge signals (See 3.19.1.3).

4.16.3.1 PRF GENERATION. Verify that only one equipment PRF cycle is established for each RF challenge (See 3.19.1.3.1).

4.16.3.2 MODE SELECTION. While performing the test of 4.16.3, set the mode selection switches to select a mode different than the mode of the challenge signal. Verify there is no PRF established for the equipment (See 3.19.1.3.2).

4.16.3.3 DECODE ENABLE. While performing the test of 4.16.3, set the decode enable switch to OFF. Verify there is no PRF established for the equipment (See 3.19.1.3.3).

4.16.4 SWEEP FREQUENCY MODE PRF. Set the equipment controls for sweep frequency operation and the range delay switches to the minimum setting. Set the trigger select switch (3.19.1) for internal trigger operation. While varying the PRT select switches from minimum to maximum settings (3.19.1.1), measure the repetition rate of the sweep sync scope trigger (3.19.6.2) for narrowband and wideband sweep operation (See 3.19.1.4).

4.16.4.1 PRF COUNTDOWN. Set the equipment controls for narrowband sweep operation and the delayed RF sweep trigger set for minimum delay. While performing the test of 4.16.4 for internal trigger operation, verify that the sweep PRF, for a PRT setting of 1000 USEC, is 500 Hz $\pm$10%. Repeat the test for wide band sweep operation with a PRT setting of 2000 USEC and verify that the resultant sweep PRF is 250 Hz $\pm$20% (See 3.19.1.4.1).

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4.16.5 RADAR TRIGGER. Using an oscilloscope, verify that a radar trigger is generated and occurs prior to the generation of the equipment's challenge video and reply signals (See 3.19.2).

4.16.6 RADAR TRIGGER CHARACTERISTICS. Measure the characteristics and timing of the output radar trigger for internal-triggered (3.19.1.1) and external-triggered (3.19.1.2) modes of operation (See 3.19.3).

4.16.7 SELECTIVE IDENTIFICATION FEATURE (SIF) PRETRIGGER. Measure the characteristics and timing of the pretrigger for internal triggered and external triggered modes of operation (See 3.19.4).

4.16.7.1 ADJUSTMENT. Measure the pretrigger timing adjustment range. Verify that the pretrigger timing is adjustable in 0.2 USEC steps (See 3.19.4.1).

4.16.8 MODE 4 GAIN TIME CONTROL (GTC) TRIGGER. Measure the characteristics and timing of the GRC trigger (See 3.19.5).

4.16.9 SCOPE TRIGGERS. Set the equipment controls for fixed frequency, internal-triggered operation. Measure the characteristics of the trigger. Repeat with the controls set for sweep operation (See 3.19.6).

4.16.9.1 FIXED FREQUENCY SCOPE TRIGGER. Measure the scope trigger timing with respect to the radar trigger (See 3.19.6.1).

4.16.9.1.1 JITTER. Measure the scope trigger jitter with respect to the radar trigger (See 3.19.6.1.1).

4.16.9.2 SWEEP SYNC SCOPE TRIGGER. Set the equipment controls for sweep operation. Measure the scope trigger timing with respect to the start of each sweep cycle (See 3.19.6.2).

4.16.9.3 DECODE SCOPE TRIGGER. Set the equipment controls to decode a Mode 3/A challenge signal. Apply a Mode 3/A RF challenge to the equipment low power input and measure the timing of the scope trigger. Repeat the test for a Mode 4 RF challenge (See 3.19.6.3).

4.16.9.4 DELAYED RF SWEEP TRIGGER. Set the equipment controls for sweep operation. Vary the front panel trigger delay control from minimum to maximum setting and measure the timing of the scope trigger with respect to the radar trigger (See 3.19.7).

4.16.10 SYSTEM TIMING.

4.16.10.1 REPLY TIMING DELAY RANGE. Set the equipment controls for internal trigger operation. Vary the delay range select switches

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(3.19.8.1.1) from minimum to maximum setting and measure the overall range of delay. Repeat the test for external trigger operation (See 3.19.8.1).

4.16.10.1.1 DELAY SELECTION. Vary the delay range select switch settings to determine that the delay is selectable in 1 USEC increments. Measure the accuracy of the delay for the following switch settings (See 3.19.8.1.1):

5 USEC	1500 USEC
10 USEC	2000 USEC
100 USEC	2500 USEC
999 USEC	2999 USEC

4.16.10.2 SIF SYSTEM TIMING.

4.16.10.2.1 TRIGGERED OPERATION. Set the equipment controls for SIF internal trigger operation. Set the reply delay switches (3.19.8.1.1) for minimum setting. Measure the time between the first bracket pulse of the SIF reply and the radar trigger (See 3.19.8.2.1).

4.16.10.2.1.1 REPLY TIMING ADJUSTMENT. While performing the test of 4.16.10.2.1, measure the timing characteristics of the reply timing adjustment (See 3.19.8.2.1.1).

4.16.10.2.2 DECODE OPERATION. Set the equipment controls for MOde 3/A decode trigger operation. Apply a Mode 3/A RF challenge to the low power input connector and measure the timing between the last challenge pulse and the first pulse of the SIF reply (See 3.19.8.2.2).

4.16.10.3 MODE 4 SYSTEM TIMING.

4.16.10.3.1 TRIGGERED OPERATION. Set the equipment controls for Mode 4 internal trigger operation. Measure the timing between the internal reference trigger and the radar trigger. Repeat for external trigger operation (See 3.19.8.3.1).

4.16.10.3.1.1 REFERENCE TRIGGER TIMING ADJUSTMENT. While performing the test of 4.16.10.3.1, measure the timing characteristics of the reference trigger timing adjustment (See 3.19.8.3.1.1).

4.16.10.3.1.2 TIMING ADJUSTMENT SYNCHRONIZATION. Set the equipment controls for Mode 4 internal trigger operation. While varying the position of the Mode 4 internal reference trigger (4.16.10.3.1.1), verify that the positioning of the Mode 4 GTC trigger (3.19.5), the Mode 4 challenge video output (3.19.10) and the Mode 4 reply (3.19.8.3.1.3) are also varied the same amount as the reference trigger (See 3.19.8.3.1.2).

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4.16.10.3.1.3 REPLY POSITION. Set the equipment controls for Mode 4 internal trigger operation. With the internal reference trigger set to its nominal position (3.19.8.3) and the range delay switches set for minimum delay, measure the time between the first pulse of the 3-pulse Mode 4 reply and the radar trigger (See 3.19.8.3.1.3).

4.16.10.3.2 DECODE OPERATION. Set the equipment controls for Mode 4 decode trigger operation. Apply a Mode 4 RF challenge to the low power input and measure the timing between the fourth interrogation sync pulse and the first pulse of the three-pulse Mode 4 reply (See 3.19.8.3.2).

4.16.11 TIMING INTERLOCK. Set the equipment controls for SIF, internal trigger operation. Set the reply delay range switches (3.19.8.1) to a setting of 100 USEC and measure the time from the radar trigger to the F_2 pulse of the SIF reply. Record the measurement results for reference. Vary the settings of the PRT switches (3.19.1.1) to a value which is 15 USEC less than the recorded measurement. Verify that the SIF reply has been inhibited. Vary the PRT switch settings to a value equal fo the reference value. The reply shall remain inhibited. Increase the setting of the PRT switches by 15 microseconds. Verify that the SIF reply output is present. Repeat the test for reply delay settings of 1000 USEC and 2000 USEC. Repeat the test for Mode 4 reply operation (See 3.19.9).

4.16.11.1 INDICATOR. While performing the tests of 4.16.11, verify that the indicator illuminates only when the reply has been inhibited (See 3.19.9.1).

4.16.12 MODE 4 CHALLENGE VIEDO OUTPUT. Count the number of output pulses and verify the pulse-to-pulse spacing (See 3.19.10).

4.16.12.1 PULSE TRAIN SPACING. Measure the spacing between the Mode 4 internal reference trigger (3.19.8.3) and the first challenge video pulse (See 3.19.10.1).

4.16.12.2 JITTER. Measure the jitter between the first challenge video pulse and the radar trigger (See 3.19.10.2).

4.16.12.3 PULSE CHARACTERISTICS. Measure the characteristics of the challenge video output pulses (See 3.19.10.3)..

4.16.13 SIF CHALLENGE VIDEO OUTPUT. Measure the characteristics of the video output pulses (See 3.19.11).

4.16.13.1 CHALLENGE VIDEO PULSE SPACING. Measure the challenge video pulse pair spacing (See 3.19.11.2).

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4.16.13.2 CHALLENGE VIDEO TIMING. Set the range delay switches for the minimum setting. Measure the time between the second pulse of the pulse pair and the first bracket pulse of the SIF reply train (3.19.11.3).

4.16.14 VIDEO CHARACTERISTICS.

4.16.14.1 SIF REPLIES. Set the equipment controls for SIF, internal trigger operation. Measure the characteristics of the SIF reply video output (See 3.19.12.1).

4.16.14.1.1 SIF NORMAL REPLIES. Set the code select switches (3.19.12.1.1.1) to code 7777. Verify that two bracket pulses and 12 information pulses are generated. Measure the spacing of each pulse with respect to the first bracket pulse (See 3.19.12.1.1).

4.16.14.1.2 CODE SELECT. Set the code select switches to a code of 0000. Verify that the SIF reply consists only of the two bracket pulses (F1 and F2). Vary the code select setting and verify the presence of SIF reply videa pulses as follows (See 3.19.12.1.1.1):

CODE	VIDEO INFORMATION PULSES
1111	A1,B1,C1,D1
2222	A2,B2,C2,D2
3333	A1,A2,B1,B2,C1,C2,D1,D2
4444	A4,B4,C4,D4
5555	A1,A4,B1,B4,C1,C4,D1,D4
6666	A2,A4,B2,B4,C2,C4,D2,D4

4.16.14.1.3 SIF IDENTIFICATION/POSITION(I/P) MODE 1 REPLY. Set the code switches to a code of 0000. Verify that two SIF pulse trains consisting only of bracket pulses are present. Measure the spacing between the Bracket 1 pulses of each train (See 3.19.12.1.2).

4.16.14.1.2 SIF I/P MODE 2/3 REPLY. Set the code switches to a code of 0000. Verify that a single SIF pulse train consisting of the two bracket pulses is present followed by a single pulse. Measure the spacing between the Bracket 1 pulse and the single pulse (See 3.19.12.1.3).

4.16.14.1.5 SIF EMERGENCY REPLY. Set the modulation select switch to EMERGENCY and the code select switches to code 0000. Verify that four sets of bracket pulses are present. Measure the spacing of the bracket pulses with respect to the first Bracket pulse of the pulse train (See 3.19.12.1.4).

4.16.14.1.6 SIF GARBLE REPLY. Set the modulation select switch to GARBLE, the code select switches to code 4000 and the substitute pulse select control (3.19.12.4) for the A4 pulse. Measure the spacing of the Garble pulse as referenced to the substitute A4 pulse in the SIF pulse train (See 3.19.12.1.5).

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4.16.14.1.6.1 GARBLE VARIABLE PULSE SPACING. While conducting the test of 4.16.14.1.6, vary the position of the A4 pulse using the substitute pulse spacing control (3.19.12.1.5.1). Measure the spacing between the Garble pulse and the substitute A4 pulse in the SIF pulse train (See 3.19.12.1.5.1).

4.16.14.2 MODE 4 REPLIES. Set the modulation select switch for a three-pulse Mode 4 reply. Measure the reply video output pulse characteristics (See 3.19.12.2).

4.16.14.2.1 THREE-PULSE REPLY SPACING. Set the modulation select switch for a three-pulse Mode 4 reply. Measure the pulse-to-pulse spacing (See 3.19.12.2.1).

4.16.14.2.2 SINGLE-PULSE REPLY POSITION. Set the modulation select switch for single-pulse Mode 4 replies. Measure the position of the single reply pulse with respect to the position of the third pulse of a three-pulse reply if a three-pulse reply was selected (See 3.19.12.2.2).

4.16.14.2.3 JAMMING REPLIES. Set the modulation select switch for three-pulse Mode 4 replies. Vary the front panel jamming selection control from minimum to maximum setting and measure the number of jamming signals provided with the normal Mode 4 reply (See 3.19.12.3).

4.16.14.2.3.1 TIMING. Measure the spacing of the jamming signals with respect to the first pulse of the normal Mode 4 reply (See 3.19.12.3.1).

4.16.14.2.3.2 JAMMING CYCLE. Verify the pattern in which the jamming signals are generated (See 3.19.12.3.1.1).

4.16.14.3 SPACING CONTROL. Set the modulation select switch to provide a normal SIF reply and a code containing the C1 and B4 pulses. Set the substitute pulse select switch to select the C1 substitute pulse. Measure the amount and accuracy of the selectable spacings for the C1 substitute pulse. Repeat the test for the following substitute pulses: B4, the second Bracket pulse (F2), and the second and third pulse of the Mode 4 three-pulse reply (See 3.19.12.4.1).

4.16.14.3.1 CONTINUOUS SPACING. Verify that the positions of the moveable reply pulses can be continuously varied from -0.8 USEC to +0.8 USEC about their nominal positions (See 3.19.12.4.2).

4.16.14.4 VARIABLE PULSE WIDTH. Measure the amount and accuracy of the selectable width settings provided by front panel controls for the reply video pulses (See 3.19.12.5).

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4.16.14.4.1 CONTINUOUSLY VARIABLE WIDTH. Measure the reply video pulse width with the continuous width control set to its minimum and maximum settings (See 3.19.12.5.1).

4.16.14.5 REPLY GATING.

4.16.14.5.1 VARIABLE INTERNAL GATING. Measure the number of replies passed and inhibited while sequencing the front panel gating pass and gating inhibit controls in accordance with the following (See 3.19.12.6.1):

PASS	INHIBIT
01	09
05	05
10	90
20	80
50	50
80	80
90	90

4.16.14.5.2 EXTERNAL GATING. Apply an external 100 millisecond wide gating signal to the equipment. Verify that output replies are present only for the duration of the external gate. Repeat the test for an external gate width of 1250 milliseconds (See 3.19.12.6.2).

4.16.15 MARKERS.

4.15.15.1 TIMING MARKERS. Verify that the markers are generated synchronously with the radar trigger (3.19.3) and are provided at a front panel connector (See 3.19.13.1).

4.16.15.1.1 MARKER CHARACTERISTICS. Measure the marker characteristics (See 3.19.3.1.1).

4.16.15.1.2 MARKER OUTPUT DURATION.

4.16.15.1.2.1 INTERNAL TRIGGER OPERATION. Set the equipment controls for internal trigger operation. Verify that the marker output is continuous (See 3.19.13.1.2.1).

4.16.15.1.2.2 EXTERNAL TRIGGER OPERATION. Set the equipment controls for external trigger operation. Apply an external trigger and measure the duration of the marker output (See 3.19.13.1.2.2).

4.16.15.2 SWEEP FREQUENCY MARKERS. Set the equipment controls for sweep frequency operation. Measure the characteristics of the frequency marker outputs (See 3.19.13.2).

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4.16.15.2.1 NARROW SWEEP BANDWIDTH. Set the equipment controls for ±5 MHz sweep frequency operation. Measure the number of markers (See 3.19.13.2.1).

4.16.15.2.2 WIDE SWEEP BANDWIDTH. Set the equipment controls for ±15 MHz sweep frequency operation. Measure the number of markers (See 3.19.13.2.2).

4.16.15.2.3 MARKER ACCURACY. Measure the accuracy of the frequency markers (See 3.19.13.2.3).

4.16.16 HIGH LEVEL RF INPUT/OUTPUT.

4.16.16.1 INPUT/OUTPUT VSWR. Measure the VSWR of both input/-output channels at the connectors (See 3.19.14.3).

4.16.16.2 RF OUTPUTS.

4.16.16.2.1 FIXED FREQUENCY. Measure the frequency of the equipment RF outputs at one of the input/output connectors (3.19.6.2) (See 3.19.14.4.1).

4.16.16.2.2 SWEEP FREQUENCY. Set the equipment controls for ±5 MHz sweep frequency operation. Verify that the sweep bandwidth of the equipment about the nominal center frequency of 1090 MHz is equal to or greater than ±5 MHz. Repeat the test for ±15 MHz sweep operation (See 3.19.14.4.2).

4.16.16.2.3 SWEEP BLANKING. Set the equipment controls for ±5 MHz sweep freuqency operation. During the sweep off time, verify that there is no modulated RF output from the equipment. Repeat the test for ±15 MHz sweep operation (See 3.19.14.4.3).

4.16.16.3 CHANNEL ATTENUATION CONTROLS. Verify that the RF output level at each input/output connector (3.19.14.1) can be varied between 0 dBm and -90 dBm (See 3.19.14.5).

4.16.16.3.1 DIFFERENTIAL ATTENUATION. Using the attenuator calibration data, verify the differential attenuation between every two consecutive 10 dB scale divisions (3.19.14.5.1) on the SUM and DIFFERENCE attenuators (See 3.19.14.5.2).

4.16.16.4 FIXED FREQUENCY RF OUTPUT LEVEL ACCURACY. Set the pulse width control to 0.45 USEC. For SUM attenuator setting of 0 dBm, -10 dBm, and -20 dBm measure the equipment's RF output level at the SUM channel RF input/output connector (3.19.14.1). Repeat the test for the DIFFERENCE channel (See 3.19.14.6).

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4.16.16.5 SWEEP FREQUENCY RF OUTPUT LEVEL FLATNESS. Set the equipment controls for ±5 MHz sweep frequency operation and a pulse width of 0.45 USEC. Measure the amplitude variation in the sweep output level over the ±5MHz bandwidth for the SUM and DIFFERENCE channel outputs. Repeat the above test for the ±15 MHz sweep bandwidth (See 3.19.14.7).

4.16.16.5.1 SWEEP OFFSET. While performing the test of 4.16.16.4, record the equipment's SUM output level for an attenuator setting of 0 dBm. Without resetting the SUM attenuator, perform the test of 4.16.16.5 and verify that the sweep output level at center frequency (1090 MHz) is within 0.5 dB of the RF output level established for the fixed frequency test. Repeat for the DIFFERENCE channel (See 3.19.14.7.1).

4.16.16.6 INTERLEAVE RF OUTPUT. Set the equipment controls for interleave operation. Verify that the equipment RF output is from the SUM input/output connector and consists of two SIF reply trains interleaved together (See 3.19.14.8).

4.16.16.6.1 SUM LEVEL. With the equipment controls set for interleave operation, verify that the SUM attenuator varies the level of the combined RF reply signal (See 3.19.14.8.1).

4.16.16.6.2 DIFFERENCE LEVEL. With the equipment controls set for interleave operation, set the SUM and DIFFERENCE attenuators to 0 dBm and measure the amplitudes of the RF output pulses. Decrease the DIFFERENCE attenuator setting to -10 dBm and verify that the DIFFERENCE channel pulse amplitude decreased 10 ±0.5 dB and that the SUM channel pulse amplitude is still 0 dBm ±1.0 dB (See 3.19.14.8.2).

4.16.16.6.3 INTERLEAVE LEVEL SET ACCURACY. While performing the test of 4.16.16.6.2, measure the interleave level set accuracy (See 3.19.14.8.3).

4.16.16.7 MODULATION.

4.16.16.7.1 INTERNAL REPLY MODULATION SIGNALS. Set the equipment front panel controls for fixed frequency operation. For each reply video signal selected by the modulation select switch, verify that a corresponding pulse-modulated RF output signal is available from the equipment's SUM and DIFFERENCE outputs (See 3.19.14.9.1.1).

4.16.16.7.2 CONTINUOUS WAVES(CW). Set the modulation select switch to CW. Measure that the power level at each of the equipment's RF outputs is 0 dBm ±1 dB using a CW power meter (See 3.19.14.9.1.2).

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4.16.16.7.3 HIGH-RATE MODULATION. Set the Modulation Select Switch to 30 KHz modulation. Verify that the RF outputs from the SUM and DIFFERENCE channels are pulse modulated at a 30 KHz ±10% rate (See 3.19.14.9.1.3).

4.16.16.7.4 EXTERNAL MODULATION. Set the Modulation Select switch of OFF. Apply an external modulating signal having characteristics as specified in 3.19.12.1 or 3.19.12.2 to the test set and observe a corresponding pulse-modulated RF output signal from the SUM channel. Set the internal modulation select switch to SIF and verify that the RF output signal is modulated by the externally applied signal and the internally generated SIF reply train (See 3.19.14.9.2).

4.16.16.7.5 INTERLEAVED MODULATION. Set the equipment controls for interleave operation. Measure the delay between corresponding pulses from the SUM RF output channel (See 3.19.14.9.3).

4.16.16.7.6 MODULATION FIDELITY. Measure the characteristics of the equipment's modulated RF output signals for SIF normal replies (3.19.12.1.1) See 3.19.14.9.4).

4.16.16.7.7 MODULATOR DELAY. Set the equipment controls for external modulation. Using a dual-trace oscilloscope, measure the delay between corresponding pulses of an external modulation signal applied to the test set and the resultant deomdulated RF output Signal (See 3.19.14.9.5).

4.16.16.7.8 CW LEAKAGE. Set the SUM attenuator setting to 0 dBm and the DIFFERENCE attenuator setting to -90 dBm. Measure the amplitude of the CW leakage signal at the DIFFERENCE channel output. Repeat the test for the SUM channel by setting the SUM attenuator to -90 dBm and the DIFFERENCE attneuator setting to 0 dBm. Measure the amplitude of the CW leakage signal at the SUM channel (See 3.19.14.9.6).

4.16.16.7.9 MODULATOR ON-OFF RATIO. Set the SUM and DIFFERENCE attenuators to -10 dBm. Measure the on-to-off ratios of the equipment's modulated RF output signals (See 3.19.14.9.7).

4.16.17 MEASUREMENT CAPABILITY. The measurement capability of the equipment shall be performed utilizing the AN/USM-281 Oscilloscope or equivalent (See 3.19.16).

4.16.17.1 DEMODULATOR SENSITIVITY.

4.16.17.1.1 HIGH LEVEL. Apply a 18 dBw (+0 dB, -0.5 dB) pulse-modulated RF signal, having the following characteristics, to the SUM channel input:

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Pulse Width 1000 Hz ±5%	0.80 ±0.1 USEC

Measure the characteristics of the demodulated video output using the AN/USM-281 Oscilloscope, or equivalent, for input frequencies of 1025 ±0.1 MHz, 1030 ±0.1 MHz and 1035 ±0.1 MHz, respectively. Repeat the test with a power input of +33 dBW (+0.5 dB, -0 dB). Repeat the above measruements for the DIFFERENCE channel input (See 3.19.16.1.1).

4.16.17.1.2 LOW LEVEL. Repeat the test of 4.16.17.1.1 with an input signal of -12 dBm (+0 dBm, -0.5 dBm) applied to the low level input (3.19.15). Measure the characteristics of the demodulated video output. Repeat the test with an input signal of +3 dBm (+0.5 dB, -0 dB) (See 3.19.16.1.2).

4.16.17.2 PULSE FIDELITY. Apply the high power RF signal, defined in 4.16.17.1.1, to the standard biased detector (with appropriate attenuation) presented in Figure 2. Measure the characteristics of the demodulated video. Apply the same signal to each of the equipment's high power input channels and measure the characteristics of the demodulated video output. Compare the results. Repeat the above tests for an RF input power level of +33 dBw ±0,5 dBw (See 3.19.16.2).

4.16.17.3 PULSE POWER MEASUREMENT CAPABILITY. Apply the RF signal defined in 4.16.17.1.1 to each of the equipment's high power input channels. Measure the power level of the input signal. For measurement of low power inputs, convert the meter indication to actual power levels using the equipment's conversion chart. Repeat the above tests for the following input signals:

	POWER LEVEL	FREQUENCY
a.	20 dBw ±0.5 dB	1025 ±0.1 MHz
b.	26 dBw ±0.5 dB	1030 ±0.1 MHz
c.	29 dBw ±0.5 dB	1030 ±0.1 MHz
d.	33 dBw ±0.5 dB	1035 ±0.1 MHz
e.	-12 dBm ±0.5 dB	1025 ±0.1 MHz
f.	0 dBm ±0.5 dB	1030 ±0.1 MHz
g.	3 dBm ±0.5 dB	1035 ±0.1 MHz
h.	30 dBm ±0.5 dB	1030 ±0.1 MHz
i.	40 dBm ±0.5 dB	1030 ±0.1 MHz

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Apply input signals e, f, g, h and i above to the equipment's low power connector using necessary attenuators (See 3.19.16.3 and 3.20.5).

4.16.17.4 FREQUENCY MEASUREMENT CAPABILITY. Set the equipment controls for frequency measurement. Apply a pulse-modulated RF signal to the equipment's SUM channel input having the following characteristics:

Pulse Width	0.80 ±0.1 USEC
PRF	1000 Hz ±5%
Power	+18 dBw (+0dB, -0.5 dB)
Input Frequency	1015 MHz, 1025 MHz, 1030 MHz, 1035 MHz, 1045 MHz, respectively

Measure the frequency of the signals. Repeat the above measruements with an input power level of 33 dBw. Repeat the above measurements for the DIFFERENCE input channel (See 3.19.16.4).

4.16.17.5 PRF MEASUREMENT. Set the test set controls for an internally triggered PRF of 9901 USEC (101 Hz). Using the equipment's measurement capabilities, measure the PRF of the internally generated replies. Measure the equipment's PRF using a frequency counter and compare the measured results. Repeat the test for PRT's of 2000 USEC (500 Hz) and 500 USEC (2000 Hz). While performing the test of 4.16.17.1.1 at an RF of 1030 MHz ±0.1 MHz, measure the PRF of the applied RF signal using the equipment's measurement capabilities (See 3.19.16.5).

4.16.17.6 SELF TEST.

4.16.17.6.1 SIGNAL ROUTING. Connect either of the equipment SUM or DIFFERENCE RF outputs to the low power input connector. Perform the tests of 4.16.17.6.1.1 and 4.16.17.6.1.2 (See 3.19.16.7.1).

4.16.17.6.1.1 PULSE FIDELITY MEASUREMENT. Measure the pulse characteristics of the equipment's modulated RF outputs. Apply the equipment's RF outputs to the standard biased detector, and measure the pulse characteristics. Compare the results of the two measurement methods (See 3.19.16.7.1.1).

4.16.17.6.1.2 POWER MEASUREMENT ACCURACY. Using the test setup of 4.16.17.6.1, measure the equipment peak power output (See 3.19.16.7.1.2).

4.16.17.6.2 BUILD-IN-TEST(BIT) FOR FREQUENCY MEASUREMENT. Set the equipment controls for sweep frequency operation. Using the front panel BIT control and the equipment frequency measurement capability, measure the frequency of the equipment's 1030 MHz BIT RF source (See 3.19.16.7.2).

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4.16.17.7 EXTERNAL VIDEO IN. Set the meter function switch for External Video In. Using a pulse generator apply an external trigger having characteristics as specified in 3.19.16.8. Verify that the PRT indication on the meter is the same as the PRF of the external video input within the tolerances of 3.19.16.5 (See 3.19.16.8).

4.16.18 POWER SUPPLY FAULT INDICATOR. In accordance with the following table, vary, in turn, each of the working voltages and verify the indicator illumination. The +1.1 volt variation from nominal test will not be required for the +5 volt working voltage (See 3.19.17):

VOLTAGE VARIATION FROM NOMINAL	INDICATOR ILLUMINATION
-1.1	YES
-0.2	NO
+0.2	NO
+1.1	YES

4.16.19 POWER SUPPLY SHORT CIRCUIT PROTECTION. In turn, short circuit each of the equipment's DC working voltages for a period of one minute each. After the shorted condition is removed, verify that the supply voltage returns to its specified level and that no damage has been done (See 3.19.18).

4.17 ACCESSORIES. (See 3.20).

4.17.1 CABLE ASSEMBLIES. Verify DC continuity for the conductors in each cable assembly. Verify that a DC short circuit condition does not exist between conductors (See 3.20.1, 3.20.2, 3.20.3 and 3.20.4).

4.17.2 VIDEO TERMINATIONS. Visually inspect the terminations to determine that the appropriate connectors have been used. Measure the DC resistance of each termination (See 3.20.6).

4.17.3 FIXED ATTENUATORS. Terminate the 15 dB attenuator with a suitable 50 ohm termination. Measure attenuator's VSWR and input impedance at frequencies of 1010, 1030, and 1050 $\pm$0.5 MHz. Remove termination and measure attenuation using the same frequencies. Repeat for the 30 dB attenuator (See 3.20.5).

4.18 OPERABILITY. The equipment shall be operated under standard test conditions (4.9). Conduct test 4.16.17.16.1.2 (See 3.22).

4.19 INTERCHANGEABILITY. Like units, assemblies, subassemblies and replaceable parts shall be checked mechanically and electrically to determine conformance to the interchangeability requirement (See 3.23).

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4.20 VISUAL AND MECHANICAL INSPECTION. Assembled equipment shall be examined for defects listed in MIL-STD-252. Prior to assembly into the equipment, printed circuit boards shall be examined for compliance with MIL-P-55110. Assemblies, modules and sections shall be examined for compliance with MIL-STD-252 (See 3.24).

4.21 QUALITY CONFORMANCE INSPECTION FOR PACKAGING. Quality Conformance for Packaging shall be in accordance with Group A, Group B, and when required (See 6.2d), the Group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

5. PACKAGING. Packaging shall be in accordance with SPI 1G00118.

6. NOTES.

6.1 INTENDED USE. Test Set, Radar AN/TPM-25A will be specifically used to align, calibrate and troubleshoot interrogator systems, such as the AN/TPX-46 and AN/TPX-50 at direct and general support levels of maintenance. The equipment provides the necessary test stimuli for maintaining interrogator systems in the field.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendments thereto.

b. First Article Inspection: Several samples of each item cited in Section 1 are generally required so that lengthy environmental and reliability tests can be completed on specified sample units while electrical performance measurements can be made on other sample units (See 3.3).

c. Marking and shipping of samples.

d. Place of final inspection.

e. Submission of the statement of treatment referenced in MIL-P-11268 paragraph titled Toxicity as soon as possible after award of contract. This statement should be submitted to the Contracting Officer.

f. Quantity of spare parts required.

6.3 GROUP C INSPECTION. Approval to ship unit(s) on order may be withheld, at the discretion of the Government, pending decision from the procuring activity until compliance with 4.5.5 is completed.

6.4 LOCATION OF OPERATIONAL INSPECTION. It is desirable that the operational inspection 4.18 be performed at a location that will minimize handling (which might cause damage to the unit(s) after this inspection.

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It is recommended that the entire lot (including the sample units) be inspected at the packaging site immediately prior to packaging.

6.5 DEFINITIONS.

6.5.1 BRANCHING. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem of filament (hyphae).

6.5.2 GROWTH COLONIZATION. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hypae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.5.3 MICROBIAL GROWTH. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.5.4 SPORULATION. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.5.5 TUBULAR GERMINATION. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5.6 PULSE SHAPE. Pulse shape parameters shall conform to Figure 1 to aid in clarifying the definitions. "Emax" is the maximum voltage amplitude of the pulse as measured from the zero axis. Points C and D on the pulse envelope are at the 90 percent emax level. Points A and F on the pulse envelope are at the 10 percent emax level. Points B and E on the pulse envelope are at 50 percent emax level.

6.5.7 PULSE RISE TIME. The rise time of a pulse is the time required for the leading edge of the pulse to rise from 10 to 90 percent of the maximum pulse amplitude. TR, as shown on Figure 1, is the rise time and is the time required for the voltage to rise between points A and C. Unless otherwise specified herein, pulse rise time shall not exceed 0.1 microsecond (USEC) nor be less than .05 microsecond.

6.5.8 PULSE DECAY TIME. The decay time of a pulse is the time required for the trailing edge of the pulse to decay from 90 to 10 percent of the maximum pulse amplitude. TD, as shown on Figure 1, is decay time and is the time required for the voltage to decay between points D and F. Unless otherwise specified herein pulse decay time shall not exceed twice the rise time or 0.20 USEC o be less than 0.05 USEC.

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6.5.9 PULSE TOP. The pulse top shall be essentially flat. The instantaneous amplitude of the pulse between the 90 percent emax points on the leading and trailing edges shall not fall below the 90 percent emax level. On Figure 1, the amplitude of any point between C and D shall not fall below the level of the line drawn through C and D.

6.5.10 PULSE DURATION. Pulse duration is the time between points B and E. which are the 50 percent maximum voltage amplitude points on the leading and trailing edges. TX, as shown on Figure 1, is the duration time.

6.5.11 DELAY OR SPACING. Delay or spacing is the time between 50 percent maximum voltage amplitude points on the leading edges of the pulses concerned, except where otherwise specified herein.

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Preparing Activity:
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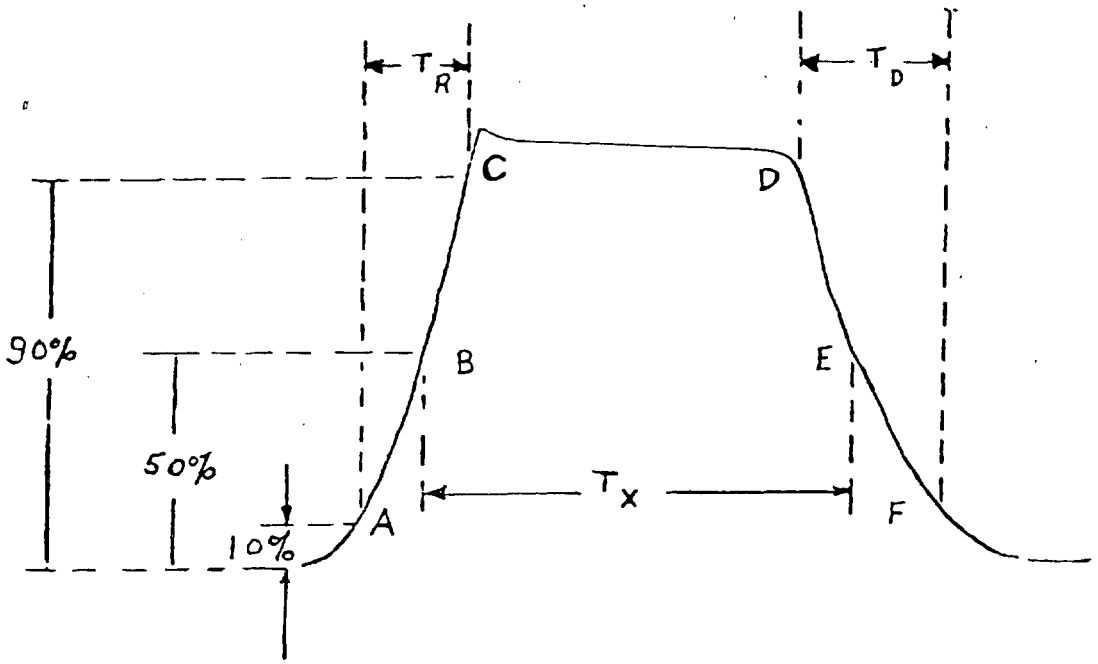


Figure 1 - Pulse Shape Characteristics

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-49057A	Test Set, IFF Interrogator AN/TPM-25A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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